Optimal Push and Pull Funding for Innovation

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Abstract

We study whether a designer should fund underprovided private sector innovations using "push funding" — such as research grants — or "pull funding" — such as prizes. We show that incentive-compatible contracts involve rewarding higher-probability-of-success firms with more in pull funding and less in push funding, and show that they can be characterized by convex functions. While push funding screens firms based on their costs of developing an innovation, pull funding selects for firms who are more likely to develop the innovation, but allows them to collect greater rents than push funding, generating a tradeoff. For a large class of distributions, the optimal contract takes a simple, "affine" form: all firms are offered the same amount of each type of funding, regardless of their type. In these cases, we show that the most cost-efficient contract always involves a positive amount of pull funding, and show precisely when a designer should also utilize a positive amount of push and pull.

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1 Introduction

Every year, governments and philanthropies spend billions of dollars funding private sector innovation. Innovation financing is a key cornerstone of modern industrial policy: in 2025, the United States government disbursed nearly $200 billion in federal funding for research and development, much of which went to the National Institutes for Health and Department of Defense (Blevins et al., 2025). In 2022, the Gates Foundation announced billions of dollars in funding for innovations that mitigate climate change, improve food security, and eradicate diseases (Gates, 2022). The AI boom is further estimated to generate up to $100 billion in annual philanthropic spending, much of which may be spent on funding lifesaving innovations (Glennerster & Rosenzweig, 2026).

External funding serves as a fundamental driver for university researchers, government contractors, and firms to develop *socially beneficial innovations* that private markets underprovide. Markets may not guarantee innovators sufficient property rights, innovations may be easy to copycat, or willingness to pay for them may be low due to buyer-side bargaining. Below, we consider three real-life examples where private sector actors cannot justify investment in innovation without external financing.

1. **Repurposed Generic Drugs**. Consider a pharmaceutical company that researches a novel use for a generic drug. The company must spend millions of dollars on clinical trials to discover a new use. However, because the drug is a generic, the firm will be forced to sell it at a competitive price, meaning profits will be slim. Budish et al. (2025) estimate the value of missing research and development in this market at $2-10 trillion.[1]

2. **Climate Resilient Crops**. There are massive social and economic returns to developing crop varieties resistant to adverse climate shocks, particularly in Sub-Saharan Africa. However, many of these crops are "open-pollen-varieties" whose seeds only need to be purchased one and can be replanted year after year. Thus, companies may only be able to profit from their innovations for a limited time (Jayne et al., 2019).

3. **Healthcare Diagnostics**. Conditions like neonatal sepsis are one of the leading causes of childhood mortality. Low-cost diagnostics that detect sepsis in neonates could save millions of lives, but current investment in similar diagnostics is low (Bansal et al., 2026). A prevailing reason is that diagnostic developers are worried about a *hold-up* problem. The primary customers for these tests are governments who will bargain prices close to marginal cost, precluding firms' ability to profit once they have invested in a diagnostic (Sen et al., 2019).

How should policymakers fund socially beneficial innovations? Most innovation is currently supported by up-front "push" funding, such as research grants. For example, a university lab may receive funding to carry out a clinical trial , but receives transfers regardless of the trial's conclusions. Another option is "pull" funding, which pays firms only upon the successful discovery

[1]The limited research that does take places in generic drug repurposing is often financed by external grants from sources like the National Institutes of Health (van der Pol et al., 2023). By contrast, *new* drugs provide firms 20 years of patent exclusivity, allowing them to reap far greater profits (U.S. Food and Drug Administration, 2020).

of innovations. These may include advance market commitments (AMCs), volume guarantees, or lump-sum prizes.[2]

This paper utilizes a mechanism design approach to answer whether policymakers should utilize push funding or pull funding. We show that policymakers should always use a *combination* of both types of funding and, in some cases, only pull funding. Moreover, we characterize the optimal mechanism for a policymaker and show that in many cases, it takes a simple form: a policymaker should offer all innovators the *same* amount in push funding and in pull funding, without needing to elicit any information about their characteristics.

In our model, a policymaker or "designer" would like to develop a socially beneficial innovation. She faces a market of firms that vary in two dimensions: their costs of costs c of research and development (R&D) and their ex-ante probability of successfully developing an innovation p. Firms' "types" (c, p) are private information to them, and are not known to the designer, who holds a prior belief over their distribution. The designer can provide ex-ante push payments to firms; pull payments that are only paid upon the successful development of the innovation; or a combination of both. She can also elicit firms' types and tailor payments to those types. Her objective is to maximize the social value unlocked with the development of the innovation, which she trades off agains the *expected cost* of paying firms to attempt the innovation. She would thus like multiple firms to "participate" in the mechanism and attempt the innovation in parallel.

We assume firms cannot embezzle funds and face no liquidity constraints. This allows us to abstract from moral hazard, limited liability, and other phenomena that would mechanically create preferences for push (or pull), and thus lets us focus on the role of *screening* in shaping the optimal mechanism.

The first part of the paper characterizes the incentive-compatible contracts — i.e. combinations of push and pull — that the designer can offer to firms who will attempt an innovation. We show that, in the absence of pull funding, the designer must provide all firms the same amount in push funding. The intuition is that since firms' costs are unobservable, firms always have an incentive to misreport their costs as higher than they actually are, meaning there is no credible way to tailor payments to types. However, with *pull funding* the designer can offer firms different amounts in push and pull funding. In particular, we show that firms who report higher probabilities of success are offered *more* in pull funding and *less* in push funding. The intuition is that pull funding is palatable to firms with high probabilities of success. The designer can guarantee truthful revelation by offering them a larger prize if they develop the innovation, but less in up-front cash.

We also provide a mathematical characterization of incentive-compatible contracts. Despite the multidimensional nature of our problem, we show that incentive-compatible mechanisms are characterized by a single "boundary" function $B(p)$, which does not depend on firms' costs and only on their reported probabilities p. $B(p)$ represents firms' indirect utility from participating

[2] An AMC resulted in the development of a vaccine for pneumococcal vaccine in the early 2010s; this included a funding pool that donor countries promised to use towards the purchase of a pneumococcal vaccine, conditional on its development (Kremer et al., 2020). Innovation prizes include those such as the Ansari XPrize for space travel (XPRIZE Foundation, 2004).

in the mechanism, sans costs. Firms enter into the mechanism and attempt innovation if and only if their costs are below that boundary, i.e. $c \leq B(p)$. This boundary function has the key property that it is *convex* in p, which is driven by the fact that push and pull funding are complements. This renders incentive-compatible mechanisms in our model slightly differently than in traditional mechanism design problems: we search for an optimal mechanism over a set of *convex* functions, rather than monotone functions. We show that if $B(p)$ is strictly convex, firms with higher probabilities of success are offered *strictly* more in pull funding (and strictly less in push). If $B(p)$ is weakly convex (i.e. affine), all firms receive the same amount in push funding and the same amount in pull funding.

Our second result is to characterize the optimal mechanism, and show that in many cases, the designer need not elicit any information from firms, and that the boundary $B(p)$ characterizing the optimal contract takes an *affine* form. As such, the designer chooses a fixed amount of push funding and pull funding, and offers any firm that applies for funding the same amount of each. Mathematically, we show that if the distribution of firms' costs is log-concave and has a decreasing density, $B(p)$ is everywhere affine. This is satisfied by many standard log-concave distributions with positive support, such as the exponential and uniform distributions. If the distribution of costs has an increasing density, $B(p)$ is weakly convex. If the distribution of costs is single-peaked, $B(p)$ is first convex and then affine.

The result is driven by two features. The first is that the very offer of pull funding inherently selects for *higher* probability of success firms. Even if all firms are offered the *same* amount in pull funding, the expected gain is higher for higher probability of success firms, who are thus more likely to select into participation. The second feature is mathematical. We show that the optimal mechanism is derived by first considering a "candidate" boundary function $b(p)$. If $b(p)$ is convex, then $B(p) = b(p)$, and $b(p)$ characterizes the optimal contract. However, in many cases, we show that $b(p)$ is concave. Thus, the designer must choose the "closest convex function" to $b(p)$, which is necessarily affine. This concept is distinct from the convexification of a curve or ironing, and is more akin to a "projection onto the set of convex functions."

We derive our optimal mechanism results first in a model with a single firm before showing that they generalize to settings with N firms, showing that the designer need not condition one firm's payments on other firms' reports or successes. We note that although policymakers may also deal with political economy constraints or prefer simple mechanisms that do not need to rely on eliciting extra information, we also provide an analytical foundation for why policymakers need not offer different firms different contracts.

Consequently, the second part of the paper investigates the case of *affine* contracts further, where different firms are all offered the same amount of each type of funding. Our third result is that the designer should always utilize a positive amount of pull funding. That is, any contract that only utilizes push funding can be improved upon by offering some pull funding. The intuition is driven by a tradeoff in rent collection between push and pull funding. Because push funding involves ex-ante payment, firms select in if their costs are below the value of that payment. Firms collect rents — driving up expected costs — but only on a single dimension. By contrast, since pull funding is only paid out upon success, firms will only participate if they have sufficiently low costs *and* sufficiently high likelihoods of success. Thus, while pull funding is more efficient at selecting in firms who are likely to succeed, firms are able to collect rents on *two dimensions*. This introduces a tradeoff. However, when the initial amount of pull funding is zero, the rents firms reap from the two mechanisms is similar, but pull's ability to *screen* on probabilities of success makes it more efficient. We note that this has a stark policy implication: the existing status quo of *grant funding* characterizing most government and philanthropic investments in innovation can be improved with outcomes-based pull funding.

Our fourth result characterizes when the designer should utilize push funding *and* pull funding, or only pull funding. We provide a sufficient condition on the distribution of costs that characterizes when the optimal mechanism involves either a strictly positive amount of push or *no* push. This condition is related to the monotone hazard rate, and characterizes the ratio of the moments of the cost distribution, which in turn represent the *relative* costs of push and pull.

Our paper finishes with some additional auxiliary results. We also compare *pure* push and *pure* pull contracts — i.e. those in which the designer *only* pays firms via push or pull (as opposed to a combination). It shows that if the expected probability at least one firm succeeds is sufficiently high *or* the number of firms in the market is small, pure push is cheaper than pure pull. If the probability is low *or* the number of firms in the market is large, pure pull is cheaper.

Finally, because pull funding often requires donors and governments to commit to funds in advance, there is also a question of what form pull funding should take to minimize *fundraising costs*. For example, should all firms be offered an individual lump-sum prize? Should a designer offer large lump-sum prizes to the first k firms that succeed and no others? We show that the form of pull funding that minimizes the amount a designer has to fundraise is given by a single pot of money that is equally divided between all successful firms.

Crucially, our results do not rely on failures in contract enforcement. For example, push funds may be susceptible to embezzlement and moral hazard[3], while pull payments may preclude participation from firms who lack up-front cash to finance research and development. Instead, we assume contracts are enforceable to highlight a fundamental difference in how these funding mechanisms *screen* firms. Firms promised push payments will attempt innovation as long as their costs of research and development are below the value of a push payment. Pull payments incentivize participation only if firms both have sufficiently low costs *and* they are sufficiently likely to develop an innovation. The combination of these two effects with the nature of rents that firms collect drives the results of our model.

Our paper is most closely connected to two other papers on push and pull. The first, Rietzke & Chen (2020), shows that the optimal mix of push and pull funding depends on a tradeoff between adverse selection and moral hazard. In their model, firms have private types determining their

[3] Kremer (2002) documents how USAID's attempt to use $60 million in push funding to develop a malaria vaccine was plagued by fund diversion and theft.

proclivity to exert effort. Firms' likelihood of success in turn depends on their effort. They show that pull funding induces effort but allows more efficient firms to collect rents, while push funding may not induce sufficient effort to induce participation. The second paper is Ridley et al. (2025), who study a general push and pull model with application to market failures in global health. Their paper has a similar setup to the previous one, and shows that "failure funding" that pays firms upon project failure may sometimes be more efficient than "pull funding." The intuition is that this reduces firms' incentives to understate their likelihoods of success, and reduces moral hazard since it is only paid out if firms exert effort.

Our paper differs from these in three fundamental ways. First, we do not consider moral hazard issues, meaning any preference for pull funding is *not* driven by firms' ability to embezzle funds or warped reporting incentives, unlike in the papers above. Second, and relatedly, we consider push and pull's fundamental ability to *screen in* different profiles of firms. Our fundamental results do not emerge because firms can collect more rents via pull, as emphasized in Rietzke & Chen (2020), or because effort is unobservable. Rather, push funding in our model screens in firms with lower costs, and pull funding screens in firms with lower ratios of costs to probabilities of success. Thus, firms collect rents from both push and pull; however, the *size* of these rents varies with the exact combination of push and pull. Finally, firms in our model do not reap any private profit beyond the transfers offered by the designer. Private profit is one of the key drivers of "failure funding" in Ridley et al. (2025)

Our predictions also differ from the papers above. Namely, while the two papers above suggest that pure push or pure pull mechanisms may be optimal, we show that any pure push contract is always improved by adding some pull. We also characterize precisely when a designer should offer *different* contracts to different firm types, or the same contract to all firms. This latter result allows us to show that a broad class of simple mechanisms is in fact optimal and easy to operationalize with public policy tools.

Our paper is more broadly connected to the literature on optimal incentives for innovation, in particular Weyl & Tirole (2012). The key difference is that while their paper is concerned with how providing market power allows regulators to *screen* projects with higher social surplus, our paper is concerned with settings where firms cannot as-is receive *any* surplus, for example, because providing market power may be unrealistic, and where the designer knows in advance what the exact innovation will be. That is, our paper applies to settings where market failures preclude market power's ability to screen, as in their paper — thus differentiating our work from a long line of literature on the use of market signals to provide innovation incentives, including Acemoglu & Linn (2004), Chari et al. (2012), Veiga & Weyl (2016)[4], and Ahlvik & van den Bijgaart (2024).

Moreover, our framework is built to answer the question of what policies — such as prizes or grants — incur the lowest expected cost for a designer, i.e. what optimal industrial policy should be in settings where governments or other funders are choosing *how* they should disburse funds.

[4]Our condition on the ratio of distributions in determining when a pure pull contract can be improved with some push is somewhat related to that in Veiga & Weyl (2016).

To this end, our paper is broadly related to the literature on the utilization of pull mechanisms to stimulate innovation in areas where the private sector does not traditionally, including studies of AMCs (Kremer et al., 2022), the direction of innovation (Bryan & Lemus, 2017), and the use of follow-on contracts to motivate procurement of efficient innovation (Che et al., 2021). Our result on prize-sharing also connects us to the literature on information disclosure in innovation contests (Halac et al., 2017).

Because pull funding screens firms on both costs and probabilities of success, our paper also contributes to the mechanism design literature on multidimensional screening. A review of the earlier literature is provided by Rochet & Stole (2003), who show how screening on multiple dimensions can link incentive contracts, in contrast to single-dimensional screening, but are oftentimes difficult to solve. Papers that solve for optimal mechanisms often rely on robustness or other techniques (see Carroll (2017), Deb & Roesler (2024), Yang (2025), and literature within). In our model, however, the two variables consisting firms' types — costs and probabilities of success — have particular interpretations. These interpretations allow us to show how a potentially complicated multidimensional screening can be reduced to a simpler form which, in our case, is given by a single convex function $B(p)$ that characterizes firm entry. The key insight, in our case, is that the *timing* of payments — whether a firm receives push payments before developing an innovation or pull funding only upon success — has a particular screening feature, and that this screening feature may lead to particularly simple optimal mechanisms. This form stands in contrast to many other problems in the literature, such as those where firms face negative externalities if competitors win a contract (e.g., through downstream competition) or consumers have multidimensional preferences over goods.

Section 2 overviews our model. Section 3 characterizes the optimal mechanism with a single firm and shows precisely when it is affine. We show how the results generalize to the N firm case at the end of the section. Section 4 zooms into the case of the affine model, showing that any pure push contract can be improved with some pull, showing when a designer may want to utilize a positive amount of push, and delivering our auxiliary results comparing pure push/pull and fundraising costs. We conclude with a brief discussion of the results.

2 Model

Setup Consider a market of firms indexed $i = 1, \ldots, N$. They are overseen by a policymaker or designer d. There are two time periods. In the first time period, each of these firms can choose to enter a project to develop a *pre-specified innovation* at cost $c_i \geq 0$. Let the variable e_i be equal to 1 if firm i undertakes the project, or "participates," and 0 otherwise. Conditional on entering, in the second time period, they succeed at developing that innovation with probability $p_i \in [0, 1]$. Similarly, let the variable s_i be equal to 1 if firm i successfully develops the innovation and 0 otherwise.

The costs c_i represent firms' ex-ante, expected research and development costs from attempting to develop an innovation. The probabilities p_i represent firms' capacity to actually develop the innovation, conditional on incurring c_i. [5] The number of prospective firms N can be thought of as representing the potential market size of innovators. Markets with high N may involve dozens of labs and private sector entities involved in the development of an innovation, while those with low N may have just a few small players developing a specific technology.

In the absence of any intervention from the mechanism designer d, we assume that firms receive no reward from successful development of the project: their utility is $-c_i$ regardless of whether the project succeeds or fails. We use bold letters throughout to indicate *profiles* of variables — e.g. $\mathbf{c} = \{c_j\}_{j=1}^n$ denotes the profile of firms' costs. We use the notation $-i$ to refer to the profile of a variable for all firms, excluding firm i.

We will refer to the tuple $t_i := (p, c)$ as a firm's *type*, and assume it is private information for firm i. It is known to (and observed only) by firm i, and unknown to other firms and the mechanism designer. We assume firms' types are drawn *independently* by Nature. That is, firms have (symmetric) priors about other firms' types, which is the same as the mechanism designer's. In particular, we assume that for each i, p_i is drawn from a continuous density g with support on $[0, 1]$ and c_i from a continuous density h with support on $\mathbb{R}^+$. We assume h is log-concave.

Designer's Mechanisms Without the mechanism designer's intervention, firms receive no utility upon successfully completing the innovation. This generates scope for the designer d to shape firm's entry decisions by choosing a mechanism that provides transfers to firms in the first or second time periods, potentially conditional on eliciting information on firms' types.

Denote $\hat{t}_i := (\hat{p}_i, \hat{c}_i)$ firm i's reported type. A *mechanism* $\mathbf{M}$ is a function that specifies entry decisions and transfers to each of the firms $i = 1, \ldots, n$ across both time periods conditional on the vector of reported types $(\hat{\mathbf{p}}, \hat{\mathbf{c}})$, decisions to undertake the project $\mathbf{e}$, and the profile of firms who succeed at the project $\mathbf{s}$. That is, a mechanism specifies entry decisions for firms $e(\hat{\mathbf{t}})$; and, for participating firms, transfers $y(\hat{\mathbf{t}}, \mathbf{e}) \in \mathbb{R}$ in the first period and $z(\hat{\mathbf{t}}, \mathbf{e}, s_{-i}) \in \mathbb{R}$ in the second period. Fixing the profile of reports $\hat{\mathbf{t}}$, firm i's ex-ante expected utility is given by:

$$e_i(\hat{\mathbf{t}})\Bigg(y(\hat{\mathbf{t}}, e_{-i}) - c_i + p_i \mathbb{E}_{s_{-i}}\big[z(\hat{\mathbf{t}}, e_{-i}, s_{-i})\big]\Bigg) \tag{1}$$

If firm i does not enter ($e_i = 0$), its utility is 0. Conditional on entering ($e_i = 1$), the firm receives a transfer $y(\hat{\mathbf{t}}, e_{-i})$, which is also a function of others' entry decisions and tye reports, less the cost c_i of undertaking the project. y can in theory be either negative or positive. In the next time period, i succeeds at the project with probability p_i, in which case it receives a transfer $z(\hat{\mathbf{t}}, e_{-i}, s_{-i})$, which it then integrates across all the different profiles of other firms s_{-i} that might succeed.

[5] The costs and probabilities can represent the aggregation of multiple stages of project development. For example, consider a drug that must pass phases A and B of a clinical trial. p_i represents the ex-ante likelihood a drug makes it through clinical trials: i.e. the probability the drug moves from A to B multiplied by the (conditional) probability the drug succeeds in phase B. c_i would be the cost of phase A plus the conditional probability of moving from phase A to B multiplied by the cost of phase B.

We refer to $y(\hat{\mathbf{t}}, e_{-i})$ as a *push payment*. If it is positive, it represents an advance payment to firms to research a problem and engage in research and development. For example, federal grant funding in the United States often takes this form. If it is negative, it represents a buy-in payment *from* firms to the designer to work on a project, such as a performance bond.

Relatedly, we refer to $z(\hat{\mathbf{t}}, e_{-i}, s_{-i})$ as a *pull payment*. It should be thought of as a prize, the expected payment from an advanced market commitment, or the total value of a subsidy or top-up payment that can be reaped *only when* a firm develops an innovation. [6]

Social Objective The mechanism designer d is concerned primarily with the development of the innovation. Formally, if at least one firm successfully enters and successfully develops the innovation, she receives a constant social surplus V. However, she trades off the development of the innovation with the *expected cost* of the mechanism, i.e. the expected cost of transfers she pays across the two time periods. For example, a government may want to incentivize the development of a vaccine to halt a pandemic. As long as firms can produce at scale, the designer simply wants to ensure that a vaccine is developed with *some* probability. Or, consider a nonprofit that wants agricultural actors to design climate resilient maize. While there may be benefits from having multiple climate resilient varieties, the nonprofit's primary objective is to ensure that farmers can benefit from at least one variety.

Given a profile $(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e})$ of reported firm types, actual probabilities of success, and entries, the expected cost of the mechanism for d is given by:

$$C(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e}) = \sum_{i=1}^{N} e_i(\hat{\mathbf{t}}) \Bigg(y(\hat{\mathbf{t}}, e_{-i}) + \mathbb{E}_{p_i}\big[p_i z(\hat{\mathbf{t}}, e_{-i}, s_{-i}) \big| \hat{p}_i, \hat{c}_i, e_i \big] \Bigg) \tag{2}$$

Formally, for each entering firm, the designer pays $y(\hat{\mathbf{t}}, e_{-i})$ in push payments. This firm may succeed or fail; the designer thus takes the expectation of p_i over the firm's report and entry decision and calculates expected push payments z.

Similarly, we also calculate the expected probability of at least one success. We denote this as $\theta(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e})$:

$$\theta(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e}) = 1 - \prod_{i=1}^{N} \mathbb{E}_{p_i}\big[1 - e_i(\hat{\mathbf{t}}) p_i | \hat{p}_i, \hat{c}_i, e_i \big] \tag{3}$$

Using this, we can write the designer d's utility, conditional on $(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e})$ as

$$V \cdot \theta(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e}) - C(\hat{\mathbf{t}}, \mathbf{p}, \mathbf{e}) \tag{4}$$

[6] It is without loss to consider pull payments that are paid out *only* upon success. If d were to pay an amount z_1 to firms upon success and z_2 upon failure, this would be equivalent to providing z_2 in push funding, paying only $z_1 - z_2$ upon success, and 0 on failure.

Designer's Problem Finally, we lay out the designer's problem. d first elicits reports of types $\hat{\mathbf{t}}$ before issuing an entry recommendation $e_i(\hat{\mathbf{t}})$, push payments $y(\hat{\mathbf{t}}, e_{-i})$, and pull payments $z(\hat{\mathbf{t}}, e_{-i}, s_{-i})$. We focus on *direct revelation* mechanisms — i.e. such that $\hat{t}_i = t_i$ — and denote deviations from firm i's type by t_i'. Note that firms do not know each others' types. Suppose $e_i(t_i, t_{-i}) = 1$. In this case, our individual-rationality and incentive-compatibility constraints can be written as:

$$\mathbb{E}_{t_{-i},s_{-i}}\Bigg[y(t_i, t_{-i}, e_{-i}(t_i, t_{-i})) - c_i + p_i z(t_i, t_{-i}, e_{-i}(t_i, t_{-i}), s_{-i})\Bigg] \geq 0 \qquad \forall t_i \text{ s.t. } e_i(t_i, t_{-i}) = 1 \tag{IR}$$

$$\mathbb{E}_{t_{-i},s_{-i}}\Bigg[y(t_i, t_{-i}, e_{-i}(t_i, t_{-i})) - c_i + p_i z(t_i, t_{-i}, e_{-i}(t_i, t_{-i}), s_{-i})\Bigg] \geq$$
$$\mathbb{E}_{t_{-i},s_{-i}}\Bigg[y(t_i', t_{-i}, e_{-i}(t_i', t_{-i})) - c_i + p_i z(t_i', t_{-i}, e_{-i}(t_i', t_{-i}), s_{-i})\Bigg] \qquad \forall t_i, t_i' \text{ s.t. } e_i(t_i, t_{-i}) = 1 \tag{IC}$$

Conditional on entry, (IR) says that firms should have an incentive to enter, taking expectation of others' types, entry decisions, and likelihoods of success. (IC) is the standard incentive compatibility constraint. If $e_i(t_i, t_{-i}) = 0$, the constraint is:

$$\mathbb{E}_{t_{-i},s_{-i}}\Bigg[y(t_i', t_{-i}, e_{-i}(t_i, t_{-i})) - c_i + p_i z(t_i', t_{-i}, e_{-i}(t_i, t_{-i}), s_{-i})\Bigg] \leq 0 \qquad \forall t_i, t_i' \text{ s.t. } e_i(t_i, t_{-i}) = 0 \tag{IR-NE}$$

Because types are drawn probabilistically, we can then write the designer's problem as

$$\max_{\mathbf{e}(\mathbf{t}), y(\mathbf{t}, e_{-i}), z(\mathbf{t}, e_{-i}, s_{-i})} \mathbb{E}_{\mathbf{t}}\big[V \cdot \theta(\mathbf{t}, \mathbf{p}, \mathbf{e}(\mathbf{t})) - C(\mathbf{t}, \mathbf{p}, \mathbf{e}(\mathbf{t}))\big] \text{ s.t. (IC), (IR), and (IR-NE)} \tag{D}$$

Discussion Our framework is meant to capture situations where firms cannot reap profits upon the successful development of an innovation, thereby precluding investment in that innovation entirely. This may be due to lack of property rights, hold-up problems in innovation procurement, political economy frictions, or other market failures. That is, our model is meant to capture precisely innovations that *depend* on industrial policy or other external transfers to come to fruition. Relatedly, this also means that there is little scope for utilizing market power or other institutional tools as a means of screening the potential value of an innovation.

To this end, a key feature of our environment is that the innovation is pre-specified by the designer, such as via a target-product-profile (TPP). For example, the designer may want firms to develop a vaccine for a specific disease that achieves a certain efficacy and safety profile. While technological agnosticism is permitted — for example, the designer may allow both mRNA and non-mRNA vaccines — our framework should *not* be thought of as a general model for funding innovation. Indeed, the benefits of many innovations may be unknown until they are discovered. Instead, our goal is to capture settings where the designer can articulate in advance the sort of product she would like firms to develop and the social value it produces.

Many of our technical assumptions are meant to focus our analysis on the role of *screening* in generating differences in expected costs between push and pull mechanisms. Firms in the model, for example, cannot embezzle a share of funds (thus creating moral hazard), which would mechanically increase the effective cost of push, thus making pull funding more attractive. Firms also do not face limited liability or liquidity constraints(e.g., because they do not have cash on hand to finance all of c_i) or discount future payments, which would make pull funding less attractive.

Finally, we assume below that firms' costs and probabilities are drawn independently. While our results go through if we allow for correlation between the two, these mechanically shift expected costs in the directions of push or pull.[7] By abstracting from all these different effects and assuming contracts are enforceable as-is, we are able to focus solely on the role of how y and z generate changes in the *composition* of firms participating and, thus, the designer's expected costs.

3 Model Solution

We dedicate the majority of this section to analyzing the optimal mechanism with $N = 1$ firms. This allows us to highlight the key intuitions of push and pull as modes of *screening* in different types of firms. In particular, we first show that the designer can only substantively condition payments to a firm based on its probability of success p.

The key result is Theorem 1. It shows that, in many cases, the optimal contract a designer can offer does not rely on eliciting any information from firms; rather, the designer offers each firm the same amount of push and pull, regardless of their probability of success p. We refer to this contract as an *affine* contract, because a firm's expected utility from accepting this contract is of the form $y + p \cdot z$ (a constant plus a scalar times a firm's probability of success p).

The precise conditions under which an affine contract is optimal depends on the distribution of firms' costs h. For cost distributions h with decreasing densities, the result holds for all $p \in [0, 1]$. For single-peaked and increasing cost distributions, it holds for an interval of p values if not for all $p \in [0, 1]$. The final subsection considers the model with $N > 1$ firms and shows that the one-firm insights apply in this setting.

3.1 Characterizing Incentive Compatibility

First, with a single firm, we rewrite the incentive-compatibility and individual rationality conditions and drop the i subscripts. Intuitively, firms' payments in this case are only a function of their reports, and firms do not have to integrate over others' entry decisions and types. To build intuition, we also expand t as $t = (p, c)$. For a firm with $e(t) = e(p, c) = 1$, the relevant constraints

[7] For example, if an increase in c_i caused a first-order stochastic shift in the distribution of p_i, our results would skew in favor of pull, while the opposite would skew in favor of push.

are:

$$y(p,c) - c + pz(p,c) \geq 0 \qquad \forall (p,c) \text{ s.t. } e(p,c) = 1 \qquad \text{(IR-1)}$$

$$y(p,c) - c + pz(p,c) \geq y(p',c') - c + pz(p',c') \qquad \forall (p,c),(p',c') \text{ s.t. } e(p,c) = 1 \qquad \text{(IC-1)}$$

We obtain a similar equation for non-entry. These equations allow us to determine the dependence of transfers on p and c.

Proposition 1. *Suppose $N = 1$. Then, $y(p,c)$ and $z(p,c)$ satisfy (IR-1) and (IC-1) if and only $y(p,c) = y(p)$, $z(p,c) = z(p)$, and there exists a convex function $B(p)$ such that $B(p) = y(p) + pz(p)$. Moreover, at all points of differentiability, we have $B(p) = z_0 + \int_0^p z(t)dt$ and $y'(p) = -pz'(p)$.*

Proposition 1 shows that the format for incentive-compatible, individually-rational mechanisms take a stark form. The expected transfers to firms $y(p,c) + pz(p,c)$ can be expressed as a convex "boundary" function $B(p)$ that does not depend on c, and which characterizes the single firm's indirect expected utility, sans costs. The proofs for this result and all results hereafter are contained in Appendix A; however, we walk through the intuition in this section.

First, we show that neither $y(p,c)$ or $z(p,c)$ can depend on c. The intuition for this comes from (IC-1): because c enters linearly into the firm's utility, conditional on entry, firms face no consequences for misreporting their costs to reap extra push funding. Because d cannot observe a firm's costs, fixing p, she cannot condition firms' payments on a firm's reported cost. We henceforth write push payments as $y(p)$ and pull payments as $z(p)$.

The intuition for $B(p)$'s convexity can be gleaned visually in Figure 1 below. The horizontal axis indicates a firm's probability of success p and the vertical axis its cost. The boundary $B(p)$ is indicated in bold. Consider, in this case, two firms with probabilities of success $p' > p$ that are both indifferent to entering, and thus lie right on the boundary $B(p)$. Thus, we have $B(p) = y(p)+pz(p)$ and $B(p') = y(p') + p'z(p')$. In particular, these two lines are tangent to $B(p)$: we have that $z(p)$ is equal to $B'(p)$ at p, and that $z(p')$ is equal to $B'(p')$.[8] Indeed, they are *supporting lines* to the convex function $B(p)$.

Suppose the type p firm reports her type as p'. This yields a utility $y(p')+pz(p')$. This is an affine function in p, where $y(p')$ represents the intercept and $z(p')$ the slope. If this firm misreported its type as p', this is equivalent to tracing out the line $y(p') + pz(p')$ and calculating its value at p, shown as a blue dotted line in the figure. Incentive compatibility necessitates that $y(p') + pz(p') \leq y(p)+pz(p) = B(p)$. That is, at p, the dotted line must be below $B(p)$. This can definitionally occur only when $B(p)$ is convex, since convex functions must lie above their supporting lines.

[8]In the case where $B(p)$ is nondifferentiable, $z(p)$ is a subgradient to $B(p)$.

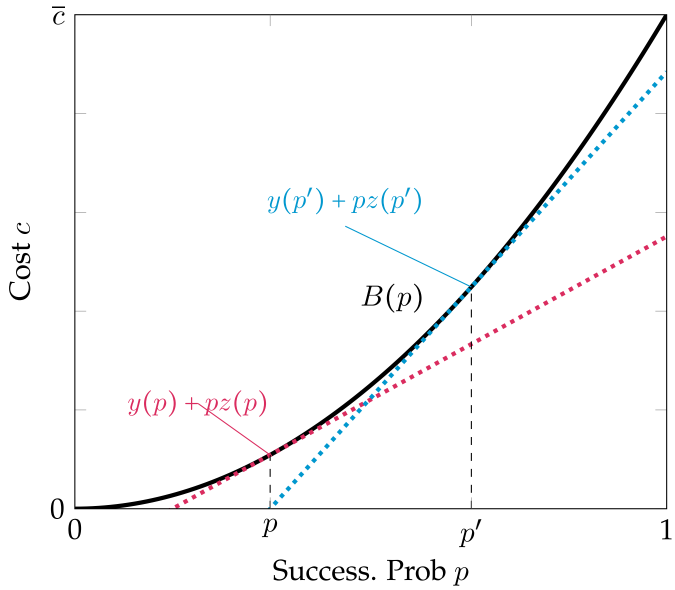


Figure 1: Boundary $B(p)$ defining incentive-compatible mechanisms

The result is an if and only if: that is, given $B(p)$ convex, we can construct a transfer $y(p)$ and $z(p)$ that satisfy this property. This insight comes from the fact that convex functions are almost everywhere twice differentiable and that, using an integration by parts argument, we have:

$$B(p) = \int z(p)dp = pz(p) - \int pz'(p)dp = pz(p) + y(p)$$

Finally, the result shows that push and pull funding are *complements*: if d would like to offer higher p firms more in pull funding, she must offer them less in push funding. In the differentiable case, we have $y'(p) = -pz'(p)$. The intuition is as follows. Suppose a firm declares it has a higher probability of success p. For a mechanism to be incentive compatible, d offers them *more* in pull funding and *less* in push funding. This form of funding is attractive only if a firm has a higher probability of success (and thus has a greater willingness to forego ex-ante payments for ex-post payments).

3.2 Designer's Problem

With our simplified results in hand, we can now move to addressing the designer's problem.

Rewriting Problem With a single firm, our problem is:

$$\max_{e(p),y(p),z(p)} \mathbb{E}_p e(p) \cdot V \cdot p - e(p)\big[y(p) + pz(p)\big] \text{ s.t. (IR-1) and (IC-1)}$$

We make two observations. First, note that the expected cost to the designer, conditional on entry, is simply $y(p) + pz(p) = \hat{B}(p)$ for some convex $\hat{B}(p)$. Second, the single firm only enters if $\hat{B}(p) \leq c$, which occurs with probability $H(\hat{B}(p))$, where H was the CDF of c. Coupled with our above results characterizing transfers in terms of $\hat{B}(p)$, we can express our problem as:

$$\max_{\hat{B}(p)} \int_0^1 \big(Vp - \hat{B}(p)\big) H(\hat{B}(p)) g(p) dp \text{ s.t. } \hat{B}(p) \text{ convex.} \tag{D-1}$$

Next, we introduce the following implicit function $b(p)$ as follows. For $p = 0$, $b(0) = 0$. For $p > 0$, we have

$$Vp = b(p) + \frac{H(b(p))}{h(b(p))} \tag{5}$$

we refer to this as the "pointwise optimum," because it solves (D-1) *pointwise* at each p value, but may or may not be convex.

The following Theorem, in turn, characterizes the solution to (D-1), which we denote $B(p)$.

Theorem 1. *Let $b(p)$ solves (5).*

- *If $h'(c) \leq 0$ for all c, then $B(p)$ is affine for all $p \in (0,1)$.*
- *If $h'(c) \geq 0$ for all c, then $B(p) = b(p)$ for all $p \in (0,1)$.*
- *If $h(c)$ is single-peaked or single-plateaued, then either $B(p)$ is affine for all p, or there exists $\overline{p} \in (0,1]$ such that $B(p) = b(p)$ for $p \leq \overline{p}$.*

Moreover, $B(p) > 0$ for all $p > 0$. Finally, if $B(p) = b(p)$, $B(p)$ is strictly convex if and only if $\frac{H(c)}{h(c)}$ is strictly concave for $b(p) = c$.

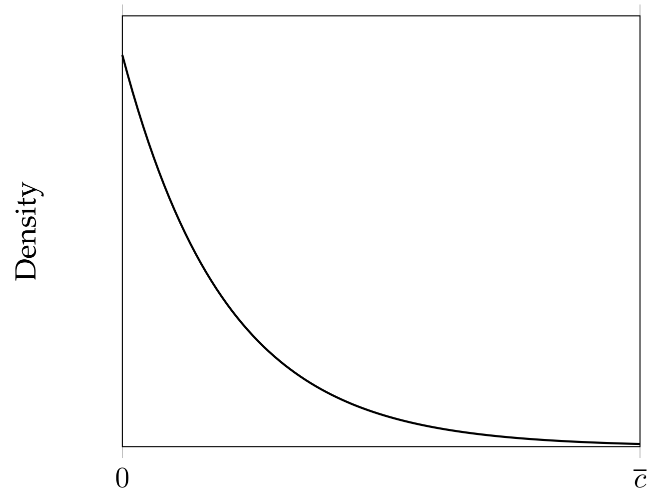
At a high level, many of the intuitions underlying Theorem 1 are similar to those from traditional mechanism design problems. In particular, the solution involves implicitly solving for a "candidate" function $b(p)$. If $b(p)$ is convex, it defines the optimal solution. However, if it is not, the optimal solution involves replacing non-convexities with line segments. Crucially, $B(p)$ is not necessarily the convexification of $b(p)$ but rather the "closest convex function" to it.

Two of the cases laid out in Theorem 1 are represented graphically in Figure 2 below. The panels on the left represent two potential distributions of costs h: decreasing and h single peaked. The panels on the right represent the optimal contract $B(p)$ in each case, given by the solid line. The "candidate" optimal contract $b(p)$ is illustrated with the dotted line.

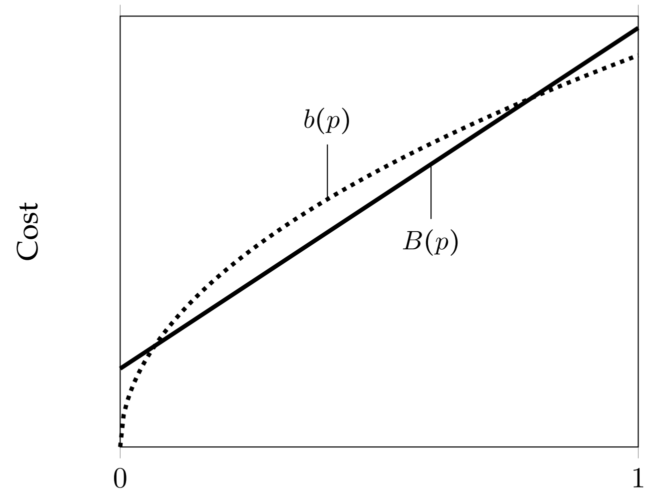
When h is decreasing, $B(p)$ is affine. The intuition is that the pointwise optimum $b(p)$ is concave. However, $B(p)$ must be convex; thus, $B(p)$ is the "closest convex function" to $b(p)$, i.e. an affine function.

The bottom panels show the case where h is single-peaked. In this case, for low p, $b(p)$ is convex. For high p, $b(p)$ is concave. Thus, although we can have $B(p) = b(p)$ for low p, we cannot have $B(p) = b(p)$ for high p. For high p, $B(p)$ thus becomes affine. In particular, if $h'(c) \geq 0$, $b(p)$ is (weakly) convex; thus, if $h(c)$ is everywhere increasing, $B(p) = b(p)$. Additionally, the proof also shows that $b(p)$ is either everywhere convex, everywhere concave, or convex and then concave. Thus, the cases laid out in Theorem 1 are exhaustive.

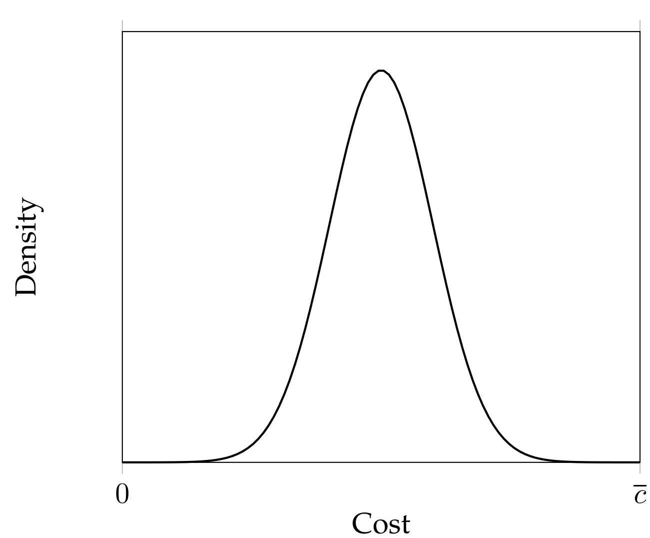
Figure 2: Different cost distributions $h(c)$ (left) and optimal contract $B(p)$ (right)



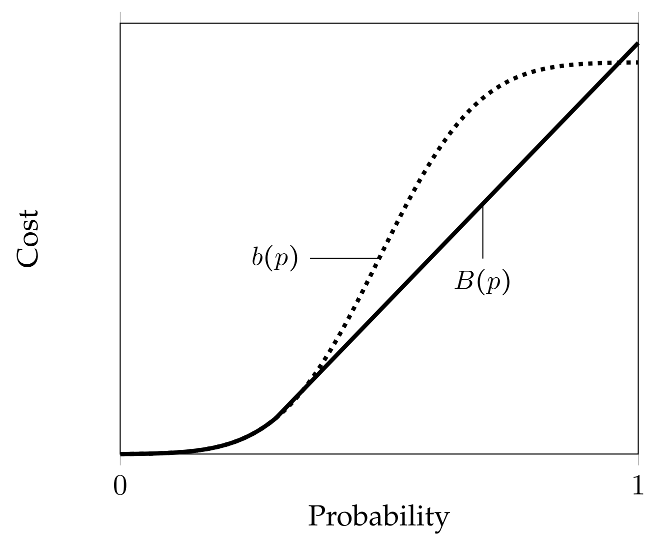
(a) Distribution of costs $h(c)$ log-concave and decreasing



(b) Optimal contract $B(p)$ (solid) vs. candidate $b(p)$ (dotted)



(c) Distribution of costs $h(c)$ log-concave and single-peaked



(d) Optimal contract $B(p)$ (solid) vs. candidate $b(p)$ (dotted)

The Theorem also notes that the concavity of $\frac{H(c)}{h(c)}$ determines the convexity of $b(p)$. In particular:

- If $h(c)$ is an exponential distribution, $\frac{H(c)}{h(c)}$ is convex, meaning $b(p)$ is concave and $B(p)$ is affine.

- If $h(c) = \frac{\alpha}{\overline{c}} c^{\alpha-1}$ on $[0, \overline{c}]$ for $\alpha \geq 1$, $\frac{H(c)}{h(c)}$ is affine, meaning $B(p)$ is affine. This includes the uniform distribution.

- If $h(c)$ is a truncated Normal or Laplace distribution with mode above 0, then $\frac{H(c)}{h(c)}$ is strictly concave then convex, meaning $B(p)$ is convex and then affine.

Whenever $B(p)$ is affine, it has a particular interpretation: we can write $B(p) = y + pz$. That is, the amount of funding each firm receives in push or pull does not depend on its type. Part of the proof also shows that whenever $B(p) \neq b(p)$, it cannot have any "kinks." Thus, whenever $B(p)$ is affine, it cannot go below the $c = 0$ line. This means that affine contracts *necessarily have positive push and pull payments*, and the designer does not need to utilize negative push payments.

In this case, the designer *does not need to elicit any information about p or c from the firm*. She always offers a push transfer of y and a pull transfer of z, regardless of what the firm reports. In this sense, the *timing* of transfers via pull itself acts as a screening mechanism. Because pull funding is inherently more attractive for firms with higher probabilities of success, they are more likely to self-select into attempting the innovation.

3.3 Model with N Firms

We now show that many of these same insights generalize when there are N firms.

IC and IR We first return to (IC) and (IR). In this case, a type i's expected revenue (i.e. indirect utility sans cost) from participating and truthfully reporting is given by

$$\mathbb{E}_{t_{-i},s_{-i}}\left[y(t_i,t_{-i},e_{-i}(t_i,t_{-i})) + p_i z(t_i,t_{-i},e_{-i}(t_i,t_{-i}),s_{-i})\right] \tag{6}$$

Note that firm i's utility integrates over all others' prospective reports. Additionally, as in the previous section, (IC) does not depend on c_i, meaning that conditional on entry, there is no way to elicit c_i. Thus, y's expected revenue can only depend on p_i, and is given (with some abuse of notation) by:

$$\tilde{B}(p_i) := \mathbb{E}_{t_{-i},s_{-i}}\left[y(p_i,t_{-i},e_{-i}(p_i,t_{-i})) + p_i z(p_i,t_{-i},e_{-i}(p_i,t_{-i}),s_{-i})\right]$$

By the same arguments as in Propositions 1, $\tilde{B}(p)$ must be convex in p.

The designer's expected utility from offering transfers that average to $\tilde{B}_i(p_i)$ for each firm is given by:

$$\left(1 - \prod_{i=1}^{N}\left(1 - \int_0^1 p_i H(\tilde{B}_i(p_i))dG(p_i)\right)\right)V - \sum_{i=1}^{N}\int_0^1 \tilde{B}_i(p_i)H(\tilde{B}_i(p_i))dG(p_i) \tag{7}$$

Crucially, note that firms and designers share common beliefs about other firms' types. Additionally, because firms and the designer are both risk neutral, it is sufficient for d to offer each firm a certain payment of $\tilde{B}_i(p)$ as opposed to a lottery over different payments. This obviates the need to utilize an auction or rely on other payment-splitting rules for entrants.

With this simplification in hand, we can state a generalized version of Theorem 1.

Theorem 2. *Suppose $\tilde{B}(p)$ solves (D). If $h'(c) \geq 0$ everywhere, there exists $\kappa \in (0,1)$ such that $\tilde{B}(p)$ and κ solve the following system*

$$h(\tilde{B}(p))\kappa V p = h(\tilde{B}(p))\tilde{B}(p) + H(\tilde{B}(p))$$
$$\kappa = \left(1 - \int_0^1 pH(\tilde{B}(p))g(p)dp\right)^{N-1}$$

If $h'(c) \leq 0$ everywhere, there exist y, z such that for all $p \in (0,1)$, $\tilde{B}(p) = y + pz$.

The insights of the Theorem are identical to the single firm case, but with the added constraint that the first-order condition defining transfers is endogeneous to the contracts being offered to other firms. The intuition is that the value of incentivizing a marginal firm in the N firm case depends on the expected likelihood that other firms enter and succeed. In particular, if N is large, the marginal value of inducing each firm to enter is small. Thus, d may be incentivized to offer each firm a less lucrative contract. But, because firms are symmetric, this decreases the probability that *all* firms enter, creating endogeneity between the contract offered to each firm through the marginal value of getting other firms to enter.

4 Analysis of Affine Contracts

For a large class of distributions — particularly those with decreasing densities — the optimal contract offered is characterized by an affine boundary $B(p)$. These contracts have the unique feature that payments do not depend on firms' types: every firm is offered the same amount of pull funding and the same amount in push funding. More broadly, in many real world settings, policymakers may be constrained in that they must offer the same amount of funding to all recipients of a grant.

Thus, in this section, we study the specific case of affine contracts with N firms, and ask when a pure push contract, a pure pull contract, or a combination is optimal. Our main result is that any "pure push" contract can be by adding some pull funding. Thus, any form of "grant-based" funding can be improved with a "pay-for-performance" element.

Additionally, we consider what form of funding is optimal when a designer can *only* choose between pure push funding and pure pull funding. We show that when the social value V of developing the innovation is small, or the number of firms N in the market is large, pure pull is more efficient than pure push. When V is large or N is small, pure push may be more efficient. Finally, subsection 4.4 shows that, among all mechanisms that involve pull, a mechanism where firms share a single pot of money involves the lowest "fundraising cost" for the designer d.

4.1 Firm Behavior

Following the notation above, we denote pure push payments y and pure pull payments z. Given a contract (y, z), firm i enters if and only if $B(p) = y + pz \geq c$. Recall from above that, in the case of an affine contract, we necessarily have $y \geq 0$; otherwise, the function $B(p)$ would necessarily have a kink. We denote $E(y, z)$ the set of firm types that enter into the mechanism, i.e. firms i such that $y + p_i z \geq c_i$.

The figure below visualizes $E(y, z)$ of firms that enter for a given y and z. Figure 3 also illustrates this visually for a mechanism where y and z are both > 0.

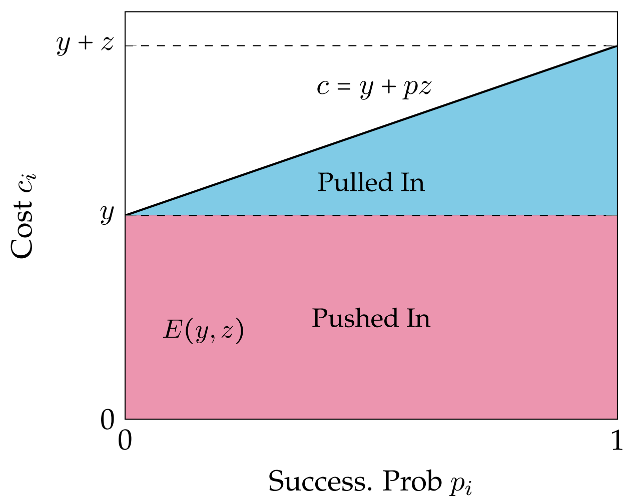


Figure 3: Set of entering firms $E(y, z)$ under affine contract (y, z)

$E(y, z)$ is given by the trapezoid in the figure. The red rectangle on the bottom represents firms whose participation is guaranteed by the push payment y. The blue triangle on the top represents firms that require both push payments y but also pull payments z.

When the size of the push payment y increases, the size of the red rectangle increases, while the size of the blue triangle stays the same; increasing push payments screens in more firms based on costs, but does not screen in firms based on their success probabilities p. When the size of the pull payment z increases, the red rectangle remains where it is, but the line $B(p) = y + pz$ capturing the edge of the blue triangle swings upwards. This screens in additional firms based on the ratio of their costs to probabilities of success. These two methods of screening will play a crucial role in our results.

We define $\sigma(y, z)$ as follows:

$$\sigma(y, z) = \int_0^1 pH(y + pz)g(p)dp.$$

σ represents the effective probability of success that emerges from a contract (y, z); it is the expected value of p for firms in $E(y, z)$ multiplied by the probability that a firm is in $E(y, z)$ to begin with. $N \cdot \sigma(y, z)$ gives the expected *number* of successes; for example, if $N = 1$, σ is the (unconditional) probability that the designer observes a success.

We can then write the affine analog of (D) as:

$$\max_{y,z\geq 0} V\big(1-(1-\sigma(y,z))^N\big) - N\int_0^1 (y+pz)H(y+pz)g(p)dp \tag{D-A}$$

4.2 Optimal Affine Contracts

We now turn to the first main result of this section, which is to show that it is always optimal for the designer to utilize some "pull" funding and, under a regularity condition, also utilize some push funding. This suggests that policymakers who may entertain flexible funding models should look to hybrid mechanisms that mix both push and pull forms of funding.

Theorem 3. *Let z^* indicate the optimal pure pull contract, i.e.*

$$z^* = \arg\max_{z\geq 0} V\big(1-(1-\sigma(0,z))^N\big) - N\int_0^1 pzH(pz)g(p)dp.$$

If $\frac{H(c)}{h(c)c}$ is weakly decreasing in c, then the optimal affine contract involves no push funding, and is given by $(0,z^*)$. If $\frac{H(c)}{h(c)c}$ is strictly increasing in c, or if H has support on $[0,\bar{c}]$ and z^* is sufficiently large, then the optimal affine contract involves a positive amount of push funding $(y>0)$.

The proof proceeds via several lemmata. Lemma 8 shows that any pure push contract can be made more cost-efficient by mixing in some pull funding. Lemma 9 shows that any pure pull contract can be made more cost-efficient by mixing in some push funding if the distribution of costs H obeys a certain constraint; sufficient conditions for these constraints are established in the consequent lemma.

Pull Improves Pure Push Contracts First, we show how any pure push contract can be improved with a small amount of pull funding. Consider first what firms are screened in with pure pull. Pure pull selects firms with smaller ratios of costs-to-probabilities of success, i.e. with lower ratios $\frac{c_i}{p_i}=\frac{c}{p}$. By contrast, push funding only selects firms with lower costs c, and does not inherently select on success probabilities p. However, by selecting in on $\frac{c}{p}$, pull allows firms to collect rents on two dimensions: both costs *and* probabilities of success. Push only allows firms to collect rents on a single dimension: costs c.

Suppose d were to utilize a *pure push* mechanism. Pull's ability to screen on p generates expected cost-savings for the designer d. Recall that $\sigma(y,z)$ represents the expected *firm-level* probability of success, integrating over the firm's probability of entry.

If d added one extra dollar of push funding at y, this would generate a marginal increase in σ of $\mathbb{E}[p]$ per dollar. However, since pull funding is only paid out (conditional on entry) with expected probability $\mathbb{E}[p]$, and in addition *selects* on higher probabilities of success if d instead added $\mathbb{E}[p]$ dollars to y, it would generate a marginal increase in σ of $\mathbb{E}[p^2]$. This logic is also borne out in Figure 4, below.

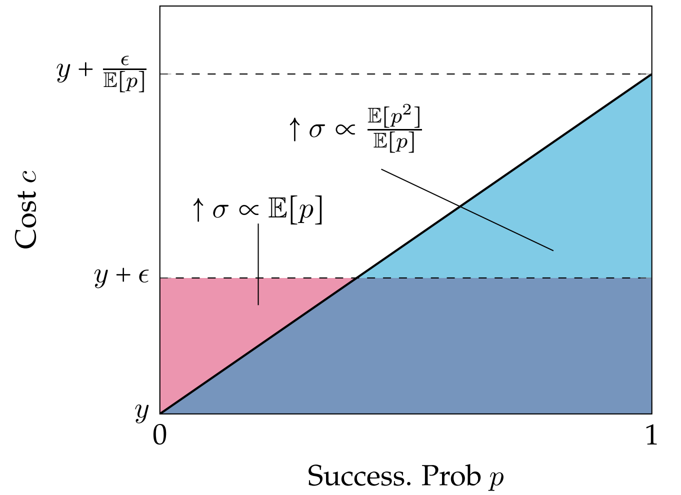


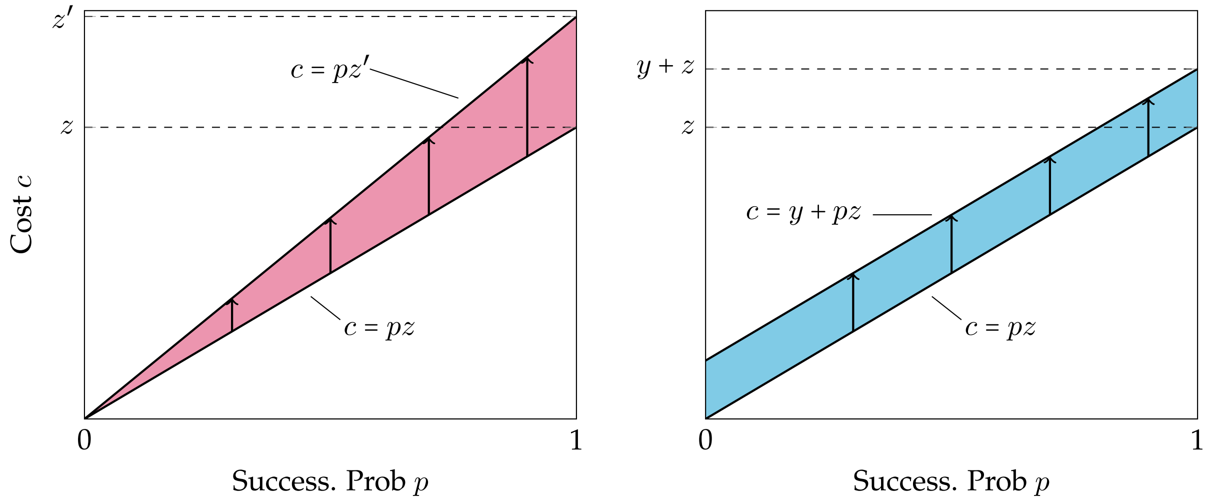
Figure 4: Improving affine pure push contract with pull

The marginal increase from y to $y+\epsilon$ leads to a marginal increase in the top of the red rectangle by ϵ. The change in σ is reflected by taking the integral reflected by the integral of p over the red rectangle, which is $\mathbb{E}[p]\epsilon$. For every dollar d spends on push, σ increases by $\mathbb{E}[p]$.

However, for every dollar d spends on pull funding, she increases σ by $\frac{\mathbb{E}[p^2]}{\mathbb{E}[p]}$. Note that pull funding is only paid out upon success. Thus, while moving one dollar to pull generates an increase of $\mathbb{E}[p^2]$, the dollar is only paid out with probability $\mathbb{E}[p]$. Thus, by incurring the same expected cost with $\frac{\epsilon}{\mathbb{E}[p]}$, the integral of σ over the blue triangle is greater than over the red area.

Push Sometimes Improves Pure Pull Contracts When beginning with a pure pull contract, the logic for introducing a positive amount of push is more subtle, and depends on properties of the distribution of costs $H(c)$. Suppose d achieved θ with a pure pull contract, i.e. with $z > 0$ and $y = 0$. A marginal increase in z, in this case, leads to an *upwards rotation* in the curve $B(p) = y+pz$. This has the positive effect of including new firms, who tend to have higher probabilities of success than under push. However, this has the negative effect that existing firms are far costlier to include, as they reap more rents. In particular, firms with relatively high costs end up seeing the greatest increases for the designer in costs of inclusion.

This is shown in the left panel of Figure 5. An increase in pull from z to z' rotates the curve $c = pz$ outwards to $c = pz'$. The red shaded area indicates the new types of firms that are screened in. The distance between the old and new curves indicates the size of *rents* that a firm of each type accrues. Crucially, while firms with low probabilities p_i accure almost no rents, firms with very high probabilities accrue more rents. Thus, while pull funding screens in more high probability of success firms, these firms also accrue the greatest increase in rents, generating a potential burden for the designer.



(a) Increasing pull from z to z'

(b) Adding push y to pull z

Figure 5: Effects of adding pull and push to affine pure pull contract

By contrast, the introduction of a push payment of z leads to an *upwards shift* in the curve $c = pz$ to $c = y + pz$. This is shown in the right panel of Figure 5, where the new firms being screened in are indicated by the blue shaded area. Unlike pull, push is less effective at selecting in firms with relatively high ps. However, the amount of rents that different firms accrue is distributed uniformly across probabilities p. Visually, the vertical distance between the new curve and old represented by the arrows is constant across firm types.

Thus, whether introducing a marginal amount of push is relatively cheaper than a marginal amount of pull depends on the curvature of costs $H(c)$ along the boundary line $c = pz$. Theorem 3 provides a set of sufficient conditions for when adding in some push can or cannot be cheaper in expectation than a pure pull contract.

If $H(c)$ is such that $\frac{H(c)}{h(c)c}$ is *strictly increasing* in c, then a pure pull contract can be improved upon by introducing a small amount of push funding. For example, the exponential distribution satisfies this property: $H(c) = 1 - e^{-\lambda c}$, with $\lambda > 0$. If $H(c)$ is such that $\frac{H(c)}{h(c)c}$ is *decreasing* in c, then a pure pull contract cannot be improved by adding a small amount of push. For example, any distribution $H(c) = \frac{1}{\bar{c}^{\alpha}} c^{\alpha}$ with $\alpha > 0$ satisfies this property, including the continuous uniform distribution. We note that under the insights of Theorems 1 and 2, the optimal contract in each case is an affine contract.

4.3 Comparing Pure Push and Pure Pull Mechanisms

Next, we answer the following question. If d could only choose between pure push and pure pull mechanisms, which one would have a lower expected cost? This constraint may reflect a realistic set of policies available to a government or funder, who must choose between one form of funding or the other.

We approach this problem by comparing the expected costs of a pure push and a pure pull mechanism that both achieve the same expected probability of at least one firm success. The following result characterizes the optimal policy.

Theorem 4. *Fix a probability of at least one success $\theta \le 1 - (1 - \mathbb{E}[p])^N$. Consider a pure pull mechanism $z(\theta)$ and a pure push mechanism $y(\theta)$ such that*

$$1 - \left(1 - \sigma(0, z(\theta))\right)^N = 1 - \left(1 - \sigma(y(\theta), 0)\right)^N = \theta$$

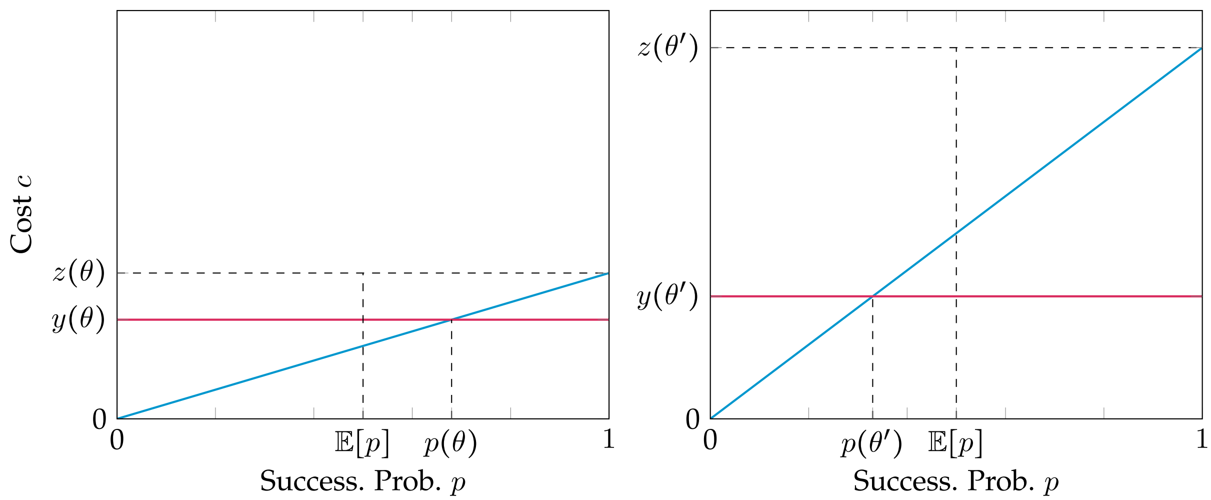
Pure pull is cheaper in expectation than pure push if and only if $\frac{y(\theta)}{\mathbb{E}[p]} \ge z(\theta)$. If θ is small, pure pull is cheaper in expectation. If θ is large, pure push is cheaper in expectation.

The comparison between pure push and pure pull is given by a simple statistic. Let $y(\theta)$ be the size of the push incentive needed to achieve a target probability θ and $z(\theta)$ the size of the pull incentive needed to achieve θ. Pure pull is cheaper in expectation if and only if

$$\frac{y(\theta)}{\mathbb{E}[p]} \ge z(\theta).$$

This inequality has an intuitive interpretation. $z(\theta)$ is the ratio $\frac{c_i}{p_i}$ of a marginal firm under pure pull. $y(\theta)$ is the value of c_i for a marginal firm under pure push, while $\mathbb{E}[p]$ is the *expected* value of p_i for that marginal firm under pure push. Whichever mechanism is cheaper depends on whichever ratio is lower.

Investigating this inequality visually provides a particularly useful characterization, shown in the two panels of Figure 6. Given the push payment $y(\theta)$, there is a firm with type $(p(\theta), y(\theta))$ such that $z(\theta) = \frac{y(\theta)}{p(\theta)}$. That is, this firm is marginal under both pure *push* and *pull*. If $p(\theta) \ge \mathbb{E}[p]$, pull is cheaper; otherwise push is cheaper. The left panel of Figure 6 shows an example where pull is cheaper: $p(\theta)$ is greater than $\mathbb{E}[p]$ at the intersection of the horizontal red line $c = y(\theta)$ (push) and the blue line $c = pz(\theta)$ (pull).



(a) Pure pull cheaper

(b) Pure push cheaper

Figure 6: Comparing pure pull and pure push contracts, different θ values

The right panel shows a different case when the designer must screen in more firms, and where push is cheaper. The intersection of $c = y(\theta')$ with the line $c = pz(\theta')$ at $p = p'$ is less than $\mathbb{E}[p]$. Whichever mechanism is cheaper will depend closely on the distributions of probabilities and costs G and H. However, Theorem 4 shows that for low θ, pure pull will always be cheaper. The designer can achieve small target probabilities of success by offering a small pull payment and select in firms with relatively high p_is, and the logic is similar to that in the proof of Theorem 3 showing that any pure push contract can be improved with some pull.

For high θ, pure push will always be cheaper. The intuition is as follows. Suppose the set of firm types were bounded above by $\bar{c}$, and the designer wanted to screen in all firm types. The cost of doing this with push would be $\bar{c}$ per firm. However, under pure pull, to screen in all firm types, even for very, very large values of z, there would always be a small mass of firms (with p close to 0) with no incentive to participate. In fact, the only way to screen in *all* firms is to send z to $+\infty$. Thus, as θ approaches $1 - (1 - \mathbb{E}[p])^N$ — which is the largest value θ can take — the cost of pure pull approaches infinity, while the cost of pure push remains finite, meaning pure push is cheaper.

A corollary of this result is that as the market of firms N grows large, pure pull begins to dominate pure push. The intuition is that for large N, the designer can "depend" on the existence of highly efficient firms, thus shrinking the set of firms she needs to screen in.

4.4 Fundraising Costs

A distinguishing feature of pull funding is that it often requires the designer to *fundraise* or secure budget items well into the future. While the amount that needs to be paid in push funding can be known relatively quickly, fundraisers may have a greater difficulty committing larger funds in the future. What is the most cost-efficient way to dole out pull funding?

In particular, recall that pull funding simply delivers an expected utility of pz to successful firms. Let P_n be the expected probability that $n \leq N$ firms succeed. Let Z_n be the amount of money in *pull funding* that a designer must pay out, conditional on n firms succeeding. The designer's fundraising cost is $\max_{i \in 1,\dots,N} Z_n$. That is, the designer must secure $\max_{i \in 1,\dots,N} Z_n$ dollars in funding.

We have so far assumed that a firm receives a certain payment z in pull funding upon success. However, we can also consider random payments to each firm that depend on the *number* of other firms that succeed, so long as each firm's expectation over payments is equal to z. Since all firms are being promised the same amount in pull rewards, if n firms succeed, their individual expected payout must be $\frac{Z_n}{n}$. Firm i's expected reward *conditional on succeeding* is then given by:

$$\sum_{n=0}^{N-1} \frac{P_n}{n+1} Z_{n+1}.$$

The designer's fundraising problem is thus

$$\min_{(Z_1,\dots,Z_N)} \max_{i \in 1,\dots,N} Z_i \text{ s.t. } \sum_{n=0}^{N-1} \frac{P_n}{n+1} Z_{n+1} = z \tag{8}$$

The solution to this fundraising problem is described in the following proposition

Proposition 2. *Consider a contract (y,z), and suppose the associated probability of at least one success is $\theta = 1 - \left(1 - \sigma(y,z)\right)^N$. The payout scheme with the lowest fundraising cost for the designer is given by a single pot of money sized at $\frac{N \cdot \sigma(y,z) \cdot z}{\theta}$, which is split equally among all successful firms. I.e., the solution to (8) is given by $Z_n = \frac{N \cdot \sigma(y,z) \cdot z}{\theta}$ for all n.*

The main consequence of the proposition is that, among all methods of doling out the (expected) payment of z to firms, the one that involves fundraising a large pot of money and paying out the entire pot to all successful firms is the cheapest.

The intuition can be gleaned as follows. Suppose that instead of a single shared pot, the designer tailored a prize of size z to all firms. The fundraising cost for this payment method would be $N \cdot z$. However, note that if $n < N$ firms succeed, the designer would have $(N - n) \cdot z > 0$ dollars left over. She can thus strictly increase successful firms' utility above z by taking those leftover $(N - n) \cdot z$ dollars and redistributing it to the n firms who did succeed. But, since she only needs to guarantee an expected value of z to all participating firms, this in turn allows her to reduce the amount she needs to fundraise in the first place.

5 Discussion

We study the problem of a designer who wants to develop a socially beneficial innovation. To stimulate private sector activity, she must offer firms external funding. She thus chooses whether to provide firms ex-ante, grant-based "push funding" y or outcomes-based "pull funding" z. Firms have two dimensional types. They vary in their costs of research and development c and their probabilities of successfully developing the innovation p, conditional on taking on those costs. Types are private information known only to individual firms, not the designer. The designer unlocks a social benefit upon the development of the innovation. She trades off the receipt of this benefit against the expected costs of motivating firms to develop the innovation, and can elicit information about firms' types as well.

We first characterize the optimal mechanism in a single-firm setting. We show that push and pull funding are complements: if the designer wants to offer firms strictly more in pull funding, she must offer them strictly less in push funding. We thus show that the optimal contract can be characterized by a convex boundary function $B(p)$; firms enter and attempt innovation if and only if their R&D costs c are below $B(p)$. Our mechanism design problem thus involves choosing the optimal convex $B(p)$ that maximizes the designer's objective. This distinguishes our problem from more standard mechanism design problems that involve maximizing surplus with respect to a monotone allocation rule.

We show how the optimal contract offered to firms varies with the distribution of firms' costs. In many cases — particularly those where cost distributions are log-concave with decreasing densities — the optimal contract takes an affine form $B(p) = y + pz$. This means that all firms are offered y in push funding and z in pull funding. We also show that in some select cases, $B(p)$ may be strictly convex, in which case push transfers $y(p)$ and pull transfers $z(p)$ vary with p (and we have $z'(p) > 0 > y'(p)$). These insights generalize to settings with N firms.

From there, we study the case of "affine contracts" where different firms are all offered the same contract. We show that any pure push contract can be improved by adding in some pull funding, and provide an additional distributional characterization for when there should be a strictly positive amount of both push and pull funding. We also study whether pure push funding dominates pure pull funding, and show that the cheapest way to operationalize pull funding is through a single pot of money that all successful firms split.

As suggested in our literature review, a key difference between our paper and the other literature on incentives for innovation is that the market as-is does not provide useful information to the designer. Put more explicitly, in the absence of transfers from the designer, there is no market for the innovation due to lack of intellectual property rights, hold-up problems, or other frictions precluding investment. Thus, the theoretical framework in this paper is also meant to be a practical guide for policymakers in governments and nonprofits who are looking to fund innovations in the face of market failures.

Our model rests on two assumptions. First, we assume away frictions due to moral hazard and limited liability. One drawback to push funding is that firms can embezzle funds or inefficiently exert effort. One drawback to pull funding is that firms may be liquidity constrained (and thus unwilling to take on risk without up-front financing), or may discount the value of profits more than a funder, thus requiring more money than they would under push. These two drawbacks cut in opposite directions, and we take no stand on their net effect in this paper to theoretically focus on these funding mechanisms' screening properties.

The second assumption is that the designer can pre-specify the fundamental properties of an innovation, such as through a target-product-profile (TPP). While pre-specifying the properties and benefits of an innovation is realistic in many settings — such as an FDA-approved vaccine or a climate resilient crop that achieves predefined performance benchmarks — it is less applicable when innovation funding is meant to stimulate general exploration of fundamental science.

With these limitations in mind, we offer a number of actionable policy takeaways.

1. The status quo for innovation funding almost entirely relies on grant-based funding. Our model suggests that this can be improved by substituting some of that grant-based funding for *pull funding* via milestone prizes, advance-market-commitments, or other outcomes-based rewards.

2. Our model suggests that, in many cases, it is sufficient to offer all firms the same contract. Thus, a policymaker need not ask firms for information about their R&D costs or likelihoods of success, nor need she run a complicated auction or other mechanism to achieve the most efficient outcome. Instead, she need only commit to offering firms an up-front grant of \$$y$ and an ex-post prize of \$$z$ to achieve the optimal mechanism.

3. As a result, pull funding obviates the need for policymakers to "pick winners." That is, a funder can rely purely on the timing of pull funding to automatically select in firms who (in firms' own view) are best positioned to succeed at developing an innovation.

4. Our model suggests that (pure) push funding may be most valuable when the necessary probability at least one firm succeeds (and thus the social value of developing the innovation) is high. That is, a heavy amount of push funding may be most valuable when time is of the essence. Nevertheless, pure push can still be improved with some pull.

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A Proofs

Proposition 1. *Suppose $N = 1$. Then, $y(p,c)$ and $z(p,c)$ satisfy (IR-1) and (IC-1) if and only $y(p,c) = y(p)$, $z(p,c) = z(p)$, and there exists a convex function $B(p)$ such that $B(p) = y(p) + pz(p)$. Moreover, at all points of differentiability, we have $B(p) = z_0 + \int_0^p z(t)dt$ and $y'(p) = -pz'(p)$.*

Proof. ($\implies$) First, our two constraints can be rearranged as:

$$\begin{aligned} &y(p,c) + pz(p,c) \geq c \\ &y(p,c) + pz(p,c) \geq y(p',c') + pz(p',c') \end{aligned}$$

Fix p. Denote $\tilde{B}(p,c) = y(p,c) + pz(p,c)$. Suppose there are two types (p,c) and (p,c') for whom IR is satisfied. Note that the IC constraint does not depend on c. Thus, if $\tilde{B}(p,c) \geq c$ and $\tilde{B}(p,c') \geq c'$, $\tilde{B}(p,c) = \tilde{B}(p,c')$.

To see that this also holds for non-participating firms, suppose $B(p,c) < c$ but that there exists c' with $B(p,c') > c$, meaning $B(p,c) > B(p,c')$. But then, this would give a type (p,c') firm an incentive to misreport its type as (p,c) to reap a larger payment, a contradiction.

Thus, we write $\tilde{B}(p,c) = B(p)$. Next, we show $B(p)$ is convex. With some abuse of notation, we write $B(p) = y(p) + pz(p)$. To see that $B(p)$ is convex, note that the IC constraint can be applied for all $p \in [0,1]$ to write $B(p)$ as

$$B(p) = \max_{p' \in [0,1]} y(p') + pz(p').$$

This is the maximum of a set of functions affine in p, and is thus convex.

Subtracting the IC constraints additionally shows that $z(p)$ is increasing in p and $y(p)$ is decreasing in p.

$$\begin{aligned} p\big(z(p) - z(p')\big) \geq y(p') - y(p) \geq p'\big(z(p) - z(p')\big) \implies (p - p')\big(z(p) - z(p')\big) \geq 0 \implies z(p') \geq z(p) \\ \implies 0 \geq p\big(z(p) - z(p')\big) \geq y(p') - y(p). \end{aligned}$$

($\impliedby$) To see the converse and the second part of the corollary, we first prove some additional features of the transfers under differentiability. Note that because z is increasing, it is almost everywhere differentiable. Let $p' > p$. Using the work above, note that we can rearrange the IC constraints as

$$\begin{aligned} p\big(z(p) - z(p')\big) \geq y(p') - y(p) \geq p'\big(z(p) - z(p')\big) \\ p\frac{z(p) - z(p')}{p' - p} \geq \frac{y(p') - y(p)}{p' - p} \geq p'\frac{z(p) - z(p')}{p' - p} \end{aligned}$$

Taking the limit as $p' \to p$ then gives

$$-pz'(p) \geq y'(p) \geq -pz'(p) \implies y'(p) = -pz'(p).$$

Using integration by parts, we additionally have that

$$y(p) + pz(p) = -\int pz'(p)dp + pz(p) = \int z(p)dp.$$

Finally, let $B(p)$ be convex. Note that $B(p)$ is almost everywhere twice-differentiable. Defining $z(p)$ as $B'(p)$ and $y(p)$ as $\int -pz'(p)dp$ gives the result. □

Theorem 1. *Let $b(p)$ solves (5).*

- *If $h'(c) \leq 0$ for all c, then $B(p)$ is affine for all $p \in (0,1)$.*
- *If $h'(c) \geq 0$ for all c, then $B(p) = b(p)$ for all $p \in (0,1)$.*
- *If $h(c)$ is single-peaked or single-plateaued, then either $B(p)$ is affine for all p, or there exists $\overline{p} \in (0,1]$ such that $B(p) = b(p)$ for $p \leq \overline{p}$.*

Moreover, $B(p) > 0$ for all $p > 0$. Finally, if $B(p) = b(p)$, $B(p)$ is strictly convex if and only if $\frac{H(c)}{h(c)}$ is strictly concave for $b(p) = c$.

The proof follows from the following lemmata.

Lemma 1. *If $\frac{H(c)}{h(c)}$ is everywhere concave in c, then $B(p) = b(p)$ for all p, where $b(p)$ solves (5).*

Proof. We first characterize the pointwise solution to the problem. Consider the following as a function of b:

$$(Vp - b)H(b).$$

Its first derivative with respect to b is ≥ 0 if and only if

$$h(b)(Vp - b) - H(b) \geq 0 \iff Vp \geq b + \frac{H(b)}{h(b)} \equiv \psi(b).$$

Note that we have an interior solution if $\psi(b)$ is increasing in b (in which case the objective is concave, and a similar argument as in Toikka (2011) applies). $\psi(b)$ is increasing in b if and only if

$$1 + \frac{h(b)^2 - H(b)h'(b)}{h(b)^2} \geq 0.$$

Since $H(b)$ is log-concave, $H(b)h'(b) \geq h(b)^2$, this is always satisfied. Thus, b is well defined, increasing in p, and constitutes an interior solution for each p.

Next, note that if $b(p)$ solves (5) and is convex, we are done. In particular, note that

$$Vp = \psi(b(p)) \implies V = \psi'(b(p))b'(p) \implies 0 = \psi''(b(p))(b'(p))^2 + \psi'(b(p))b''(p)$$
$$\implies b''(p) = -\frac{\psi''(b(p))(b'(p))^2}{\psi'(b(p))}$$

Since $\psi' \geq 0$, we have that $\mathrm{sign}(b''(p)) = -\mathrm{sign}(\psi''(b(p)))$. But note that $\psi''(b)$ is simply given by $\left(\frac{H(b)}{h(b)}\right)''$. Thus, if $\frac{H(c)}{h(c)}$ is everywhere concave in c, $b(p)$ solves $B(p)$, and we are done. □

Lemma 2. *If $h'(c) < 0$, $\frac{H(c)}{h(c)}$ is strictly convex.*

Proof. Note that $\frac{H(c)}{h(c)}$ is strictly convex if and only if

$$\left(\frac{-H(c)h'(c)}{h(c)^2}\right)' = -h(c)\frac{h'(c)}{h(c)^2} - h''(c)\frac{H(c)}{h(c)^2} + 2H(c)\frac{h'(c)^2}{h(c)^3} > 0$$
$$H(c)\big(2h'(c)^2 - h''(c)h(c)\big) > h'(c)h(c)^2$$

Because h is log-concave, $h''(c)h(c) - h'(c)^2 \leq 0 \implies 2h'(c)^2 - h''(c)h(c) \geq 0$. This means the left-hand side is always positive. Thus, a sufficient condition for the inequality to hold (i.e. for $\frac{H(c)}{h(c)}$ to be *convex*) is that $h'(c) < 0$. □

Lemma 3. *$\frac{H(c)}{h(c)}$ is weakly convex whenever $h'(c) > 0$.*

Proof. First, recall that log-concave distributions are always single-peaked or single-plateaued. Thus, consider log-concave distributions with regions of increasing density. Let $m(h) = \max\{c : h'(c) \geq 0\}$. Suppose $h'(c) \geq 0$ on a region $[0, m(h)]$. We construct the following PDF on the region $[0, 2m(h)]$.

$$\overline{h}(c) = \begin{cases} \frac{h(c)}{2H(m(h))} & c \leq m(h) \\ \frac{h(2m-c)}{2H(m(h))} & c \geq m(h) \end{cases}$$

Note that $\overline{h}$ is a well-defined distribution. We claim that $\overline{h}$ is log-concave on $[0, 2m(h)]$. Log-concavity is immediate on $[0, m(h)]$. To see log concavity on $[0, m(h)]$ note that $\overline{h}'(2m - c) = -h(2m - c)$, meaning

$$\left(\log\left(\overline{h}(c)\right)\right)'' = \left(\frac{\overline{h}'(c)}{\overline{h}(c)}\right)' = \frac{\overline{h}''(c)\overline{h}(c) - (\overline{h}'(c))^2}{(\overline{h}(c))^2} = \frac{h''(2m-c)h(2m-c) - h'(2m-c)^2}{(h(2m-c))^2}.$$

Thus, $\overline{h}(c)$ is symmetric around $m = m(h)$, log-concave, increasing for low h, and decreasing for high h. Let $\overline{H}(c)$ indicate the CDF of this distribution. By the previous lemma, note that $\frac{\overline{H}(c)}{\overline{h}(c)}$ is convex for $c > m$, since $h'(c) \leq 0$ for $c > m$. Next, consider the function $\phi(c) = \frac{\overline{H}(c+m)}{\overline{H}(c+m)} - 1/2$. Because $\overline{h}$ is symmetric about m, ϕ is odd, ϕ' is even, and ϕ'' is odd. Because $\phi'' \geq 0$ above m by the previous lemma, $\phi'' \leq 0$ below m. Thus, $\frac{\overline{H}(c+m)}{\overline{h}(c+m)}$ must be concave for $c \leq 0$ and convex for $c \geq 0$; normalizing to $\frac{H(c)}{h(c)}$ gives the result. □

The key insight of this lemma is that $b(p)$ is either everywhere concave, everywhere convex, or convex and then concave. These are the only potential shapes of $b(p)$ we need to consider.

Lemma 4. *Suppose there exists an open interval $O \subseteq [0,1]$ such that $B(p) \neq b(p)$. Then, $B(p)$ is equal to an affine function on O.*

Proof. Note that because $B(p)$ is convex, it is almost everywhere twice-differentiable. In particular, this means that we can write its solution as an optimal control problem. Our state variables are

$$B(p) \text{ and } s(p) = \dot{B}(p) \implies \dot{B}(p) = s(p) \text{ and } \dot{s}(p) = u(p),$$

where $u(p) = \dot{s}(p) = B''(p)$ is our control variable. The constraint on the control is that $\dot{s}(p) = u(p) \geq 0$ at all points of differentiability. Let $\lambda_B(p)$ be the co-state on $\dot{B}(p) = s(p)$ and $\lambda_s(p)$ be the co-state on $\dot{s}(p) = u(p)$. Our Hamiltonian is thus

$$\mathcal{H}\big(B(p), s(p), u(p), \lambda(p), p\big) = \big(Vp - B(p)\big)H(B(p))g(p) + \lambda_B(p)s(p) + \lambda_s(p)u(p).$$

Note that we are choosing $u(p) \geq 0$ to maximize $\mathcal{H}$; this amounts to maximizing $\lambda_s(p)u(p)$, which also means we must have $\lambda_s(p) \leq 0$.

By Pontryagin's Maximum Principle, we have

$$-\dot{\lambda}_B(p) = h(B(p))\big(Vp - B(p)\big)g(p) - H(b(p))g(p) \tag{9}$$

$$-\dot{\lambda}_s(p) = \lambda_B(p) \tag{10}$$

Now, let O be as in the hypothesis. Note that we must either have $b(p) > B(p)$ or $b(p) < B(p)$ on O; suppose that the former holds. This means the right hand side of (9) is strictly positive, meaning $-\dot{\lambda}(p) > 0$ on all of O. Hence, $\lambda_B(p)$ is strictly increasing on O, meaning $-\dot{\lambda}_s(p)$ is non-constant.

Recall that when $u(p_0) > 0$, for $\lambda_s(p)u(p)$ to have a well-defined maximum, we must have $\lambda_s(p_0) = 0$. But then, because λ_S is strictly increasing on O, $\lambda_S(p_0 + \epsilon) > 0$, a contradiction. An identical argument applies for the case where $B(p) > b(p)$. Thus, at all points of differentiability, whenever $B(p) \neq b(p)$, we must necessarily have $B(p)$ affine.

This leaves open the possibility that $B(p)$ is composed of multiple affine line segments glued together at kinks. However, the same argument can be applied by taking a smooth approximiation to $B(p)$ in a small neighborhood of its kinks, which will necessarily be strictly convex at the kink. □

Lemma 5. *Suppose there exists $p^\dagger \in (0,1)$ such that $b(p)$ is convex on $[0, p^\dagger)$ and concave (strictly at least one point) on $[p^\dagger, 1]$. Then, $B(p)$ is equal to a single affine function on all of $[p^\dagger, 1]$. Moreover, $B(p)$ equals*

$b(p)$ at least once on $[p^\dagger, 1]$.

Proof. Let $P = [0, p^\dagger]$. Let $O = [p^\dagger, 1]$. Note that on O, $b(p)$ can equal $B(p)$ at most twice; if it intersected at least three times, this would contradict $B(p)$ being convex.

Next, we claim that $B(p) = b(p)$ at least once on O. Suppose by contradiction that $B(p) \neq b(p)$ on all of O. By the previous lemma, $B(p)$ is then necessarily equal to a single affine function on all of O. Moreover, we must have $B(p) = b(p)$ at least once on P; if not, we could generate an improvement by shifting $B(p)$ up or down to match the pointwise optimum $b(p)$ at least once. Let $\overline{p}$ be the right-most point on P such that $B(\overline{p}) = b(\overline{p})$.

There are two cases to consider here. First, suppose $B'(\overline{p}) \leq b'(\overline{p})$, in which case, $B(p)$ is "shallower" than $b(p)$. However, we can construct an improvement by fixing $B(p)$ at $\overline{p}$ but rotating its right portion upwards until it hits $b(p)$ at least once.

Second, suppose $B'(\overline{p}) > b'(\overline{p})$, in which case, $B(p)$ is "steeper" than $b(p)$. We then have $B(p) > b(p)$ for all $p \geq \overline{p}$. We construct an improvement as follows. Let $\mathrm{conv}(b(p))$ denote the convexifictation of $b(p)$. Note that there exists p_c such that $\mathrm{conv}(b(p)) = b(p)$ for all $p \leq p^c$ and $\mathrm{conv}(b(p)) < b(p)$ for all $p \in (p^c, 1)$. By the hypothesis, we must necessarily have $\overline{p} < p^c$, meaning $b'(\overline{p}) < b'(p^c)$. In particular, this means that a line segment with slope $b'(\overline{p})$ starting from $\overline{p}$ lies strictly below $B(p)$. Thus, we can reduce the steepness of the slope $B(p)$ downwards towards $b'(\overline{p})$. By the Intermediate Value Theorem, it must touch $b(p)$ at least once on O.

Thus, $B(p)$ must intersect $b(p)$ once or twice on O. Let p_0 be the minimum of these intersection points: $\min\{p : B(p) = b(p)\}$. We show that $B(p)$ cannot have a kink at p_0 for $p_0 < 1$. Thus, suppose by contradiction that $B(p)$ has a kink at p_0, i.e. $\lim_{p \to p_0^-} B'(p) < \lim_{p \to p_0^+} B'(p)$.

First, consider the case where for all $p < p_0$, $B(p) > b(p)$. In this case, we can construct an improvement by marginally lowering the slope of $B(p)$ to be closer to the pointwise optimum $b(p)$, contradicting that $B(p)$ was the optimum. Thus, as above, there must exist $\overline{p} \in (0, p_0)$ such that $B(\overline{p}) = b(\overline{p})$. By the Intermediate Value Theorem, we must necessarily have that $B(p) > b(p)$ for all $p \in (\overline{p}, p_0)$. By the previous lemma, $B(p)$ is affine on this segment with slope $\frac{b(p_0) - b(\overline{p})}{p_0 - \overline{p}}$. Moreover, we must have $\overline{p} < p^\dagger$; otherwise, this would contradict the hypothesis that p_0 was the leftmost point of intersection on $[p^\dagger, 1]$. But in this case, note that we can marignally lower the slope of $B(p)$ on $[\overline{p}, p^\dagger]$ so that it intersects $b(p)$ at a point $\overline{p}' > \overline{p}$ and allow $B(p) = b(p)$ on $[\overline{p}, \overline{p}']$. This contradicts the fact that $B(p)$ was the original optimum.

In the case where $B(p_0)$ is the only intersection with $b(p)$ on O, we are done. However, now consider the case where it intersects twice on O, and let p_0' be the higher of those two points. We rule out a kink at p_0' as well. On $[p_0, p_0']$, note that $B(p)$ is a line segment with slope $b' = \frac{b(p_0') - b(p_0)}{p_0' - p_0}$. By the Mean Value Theorem, there exists $p_1 \in (p_0, p_0')$ with $b'(p_1) = b'$. By convexity of B and concavity of b, for a kink to occur at p_0', we must have that the slope of $B(p)$ to the right of p_0' is strictly larger than p'. This means that $B(p) > b(p)$ to the left of p_0'. But note that we can also improve this by lowering $B(p)$ to the right of p_0' by making it shallower until it equals b', bringing it closer to the pointwise optimum $b(p)$, thereby constructing an improvement

Hence, $B(p)$ is affine with no kinks on $[p^\dagger, 1]$. □

Lemma 6. *Suppose there exists $p^\dagger \in (0,1)$ such that $b(p)$ is convex on $[0,p^\dagger)$ and concave (strictly at least one point) on $[p^\dagger,1]$. Then, either $B(p)$ is affine on all of $[0,1]$, or there exists $\overline{p} \in (0,p^\dagger)$ such that $B(p) = b(p)$ for all $p \leq \overline{p}$.*

Proof. Again let $P = [0,p^\dagger]$ and $O = [p^\dagger,1]$. If $B(p)$ intersects $b(p)$ on O but not on P, we are done. If $B(p)$ intersects $b(p)$ on P at a point $\overline{p}$ and $B'(\overline{p}) < b'(\overline{p})$, we are also done, and $B(p)$ is everywhere affine. Finally, if $B'(\overline{p}) \geq b'(\overline{p})$, allowing $B(p) = b(p)$ for $p \geq \overline{p}$ allows $B(p)$ to be convex and always achieve the pointwise optimum above p. □

Lemma 7. *$B(p) > 0$ for all $p > 0$.*

Proof. First, note that by the first-order condition (5) used to define $b(p)$, $b(0) = 0$. Additionally, $b(p)$ is strictly increasing. If $B(p) = b(p)$ we are done.

Otherwise, suppose there existed some $p_0 > 0$ such that $B(p_0) = 0$. Note that $B(p_0) \neq b(p_0)$. Thus, by a previous lemma, $B(p)$ is equal to an affine function in a neighborhood of p_0. Note that because $B(p)$ is affine here, and H has positive support, this means that either $B(p)$ has a kink at p_0 or $B(p) = 0$ for all p. The former contradicts a previous lemma, and the latter can be improved by a positively-sloped linear function $B(p) = pz$ for some $z > 0$ that is closer to $b(p)$. □

Theorem 2. *Suppose $\tilde{B}(p)$ solves (D). If $h'(c) \geq 0$ everywhere, there exists $\kappa \in (0,1)$ such that $\tilde{B}(p)$ and κ solve the following system*

$$h(\tilde{B}(p))\kappa V p = h(\tilde{B}(p))\tilde{B}(p) + H(\tilde{B}(p))$$

$$\kappa = \left(1 - \int_0^1 pH(\tilde{B}(p))g(p)dp\right)^{N-1}$$

If $h'(c) \leq 0$ everywhere, there exist y, z such that for all $p \in (0,1)$, $\tilde{B}(p) = y + pz$.

Proof. Many of the arguments of this proof are the same as in the proof of Theorem 1. In particular, suppose d offers firm i a transfer of b when her type is p. The implicit formula defining b is given by

$$\kappa V p = b + \frac{H(b)}{h(b)}$$

where $\kappa \in (0,1)$ is an (endogeneous) indeterminate constant equal to $\left(1 - \int_0^1 pH(\tilde{B}(p))g(p)dp\right)^{N-1}$ in equlibrium. Defining ψ as in the proof of Theorem 1, we have

$$\kappa V p = \psi(b).$$

Define $\tilde{b}(p)$ as the solution to the above. Notably, monotonicity and convexity of $\tilde{b}(p)$ do not depend on κ, only on $\frac{H(b)}{h(b)}$. In the case where $\frac{H(b)}{h(b)}$ is everywhere concave, κ and $\tilde{b}$ are then given by the solution to the simultaneous system:

$$\begin{aligned}\tilde{b}(p) &= \psi^{-1}\big(\kappa V p\big)\\ \kappa &= \Big(1 - \int_0^1 pH(\tilde{b}(p))g(p)dp\Big)^{N-1}.\end{aligned}$$

In the case where $\frac{H(b)}{h(b)}$ is everywhere convex, for each κ, $\tilde{b}(p)$ will be concave. Thus, by the same argument as in Theorem 1, the optimal solution is necessarily affine on $(0,1)$.

□

Theorem 3. *Let z^* indicate the optimal pure pull contract, i.e.*

$$z^* = \arg\max_{z\geq 0} V\big(1-(1-\sigma(0,z))^N\big) - N\int_0^1 pzH(pz)g(p)dp.$$

If $\frac{H(c)}{h(c)c}$ is weakly decreasing in c, then the optimal affine contract involves no push funding, and is given by $(0,z^*)$. If $\frac{H(c)}{h(c)c}$ is strictly increasing in c, or if H has support on $[0,\overline{c}]$ and z^* is sufficiently large, then the optimal affine contract involves a positive amount of push funding $(y>0)$.

Proof. The proof of this result follows from the following three lemmata. □

Lemma 8. *Any pure push contract can be improved upon with some pull. That is, any contract (y,z) with $y>0$ and $z=0$ can be improved by a contract with $y>0$.*

Proof. First, we write the derivative of $\sigma(y,z)$ with respect to y and z:

$$\begin{aligned}\sigma_y &= \frac{d}{dy}\int_0^1 pg(p)H(y+pz)dp = \int_0^1 pg(p)h(y+pz)dp\\ \sigma_z &= \frac{d}{dz}\int_0^1 pg(p)H(y+pz)dp = \int_0^1 p^2g(p)h(y+pz)dp\end{aligned}$$

Next, recall the designer's problem:

$$\max_{y,z\geq 0} V\big(1-(1-\sigma(y,z))^N\big) - N\int_0^1 (y+pz)H(y+pz)g(p)dp \tag{D-A}$$

The first derivatives of (D-A) with respect to y and z are:

$$\begin{aligned}&[y]: \quad N\cdot V\cdot(1-\sigma(y,z))^{N-1}\sigma_y - N\int_0^1 H(y+pz)g(p)dp - N\int_0^1 (y+pz)h(y+pz)g(p)dp\\ &[z]: \quad N\cdot V\cdot(1-\sigma(y,z))^{N-1}\sigma_z - N\int_0^1 pH(y+pz)g(p)dp - N\int_0^1 p(y+pz)h(y+pz)g(p)dp\end{aligned}$$

Now, suppose by contradiction that an optimal affine contract had $y>0$ and $z=0$. Then, it must be the case that

$$V \cdot (1-\sigma(y,0))^{N-1}\sigma_y = \int_0^1 H(y)g(p)dp + \int_0^1 yh(y)g(p)dp = H(y) + yh(y)$$
$$V \cdot (1-\sigma(y,0))^{N-1}\sigma_z \leq \int_0^1 pH(y)g(p)dp + \int_0^1 pyh(y)g(p)dp = H(y)\mathbb{E}[p] + yh(y)\mathbb{E}[p]$$

I.e., the first order condition with respect to y binds with equality and the first order condition with respect to z is negative at 0. In this case, note that

$$\sigma_y = \int_0^1 pg(p)h(y)dp = h(y)\mathbb{E}[p]$$
$$\sigma_z = \int_0^1 p^2 g(p)h(y)dp = h(y)\mathbb{E}[p^2].$$

Thus, rearranging the FOCs gives:

$$\frac{H(y)\mathbb{E}[p] + yh(y)\mathbb{E}[p]}{h(y)\mathbb{E}[p^2]} \geq V \cdot (1-\sigma(y,0))^N = \frac{H(y) + yh(y)}{h(y)\mathbb{E}[p]}$$
$$\implies \mathbb{E}[p]^2 \geq \mathbb{E}[p^2]$$

which is a contradiction for h continuous. Hence, no contract $(y,0)$ can be optimal in the space of affine contracts. □

Lemma 9. *Any pure pull contract can be improved upon with some push only if the expression*

$$\frac{\int_0^1 pg(p)H(pz)dp}{\int_0^1 p^2 g(p)h(pz)dp} - \frac{\int_0^1 g(p)H(pz)dp}{\int_0^1 pg(p)h(pz)dp} \tag{11}$$

is > 0. That is, any contract (y,z) with $y=0$ and $z>0$ can be improved upon only if the equation above holds.

Proof. Consider a contract with $y=0$ and $z>0$. Using the argument above, the derivatives of the objective with respect to y and z at these points are ≥ 0 if and only if

$$[y]: \qquad V \cdot (1-\sigma(0,z))^{N-1}\sigma_y - \int_0^1 H(pz)g(p)dp - z\int_0^1 pg(p)h(pz)dp \geq 0$$
$$[z]: \qquad V \cdot (1-\sigma(0,z))^{N-1}\sigma_z - \int_0^1 pg(p)H(pz)dp - z\int_0^1 p^2 g(p)h(pz)dp \geq 0$$

We also have

$$\sigma_y = \int_0^1 pg(p)h(pz)dp$$
$$\sigma_z = \int_0^1 p^2 g(p)h(pz)dp$$

Utilizing a similar argument as in the previous lemma, if $(0,z)$ is an optimum, we must have that the derivative with respect to z is 0 and with respect to y is ≤ 0. Using the above, we have:

$$\frac{\int_0^1 pg(p)H(pz)dp + z\int_0^1 p^2 g(p)h(pz)dp}{\int_0^1 p^2 g(p)h(pz)dp} \leq \frac{\int_0^1 g(p)H(pz)dp + z\int_0^1 pg(p)h(pz)dp}{\int_0^1 pg(p)h(pz)dp}$$

$$\frac{\int_0^1 pg(p)H(pz)dp}{\int_0^1 p^2 g(p)h(pz)dp} \leq \frac{\int_0^1 g(p)H(pz)dp}{\int_0^1 pg(p)h(pz)dp}.$$

This means that $y > 0$ can improve the contract if

$$\frac{\int_0^1 pg(p)H(pz)dp}{\int_0^1 p^2 g(p)h(pz)dp} - \frac{\int_0^1 g(p)H(pz)dp}{\int_0^1 pg(p)h(pz)dp} \tag{11}$$

is > 0. □

Lemma 10. *The inequality*

$$\frac{\int_0^1 pg(p)H(pz)dp}{\int_0^1 p^2 g(p)h(pz)dp} - \frac{\int_0^1 g(p)H(pz)dp}{\int_0^1 pg(p)h(pz)dp} \tag{11}$$

is ≤ 0 *if* $\frac{H(c)}{h(c)c}$ *is weakly decreasing in* c*. It is* > 0 *if* $\frac{H(c)}{h(c)c}$ *is strictly increasing in* c*, or if* H *has support on* $[0, \overline{c}]$ *and* z *is sufficiently large.*

Proof. We rewrite the inequality in (11) as

$$\left(\int_0^1 pg(p)H(pz)dp\right)\left(\int_0^1 pg(p)h(pz)dp\right) > \left(\int_0^1 g(p)H(pz)dp\right)\left(\int_0^1 p^2 g(p)h(pz)dp\right)$$

By Fubini's Theorem, this can be rewritten as

$$\int_0^1\int_0^1 pg(p)H(pz)qg(q)h(qz)dqdp > \int_0^1\int_0^1 g(p)H(pz)q^2 g(q)h(qz)dqdp$$

$$\int_0^1\int_0^1 g(p)g(q)H(pz)h(qz)[pq - q^2]dqdp > 0$$

$$\int_0^1\int_0^1 g(p)g(q)H(pz)h(qz)q(p-q)dqdp > 0$$

Dividing by two and switching the order of integration allows us to rewrite the LHS as:

$$\frac{1}{2}\int_0^1\int_0^1 [g(p)g(q)H(pz)h(qz)q(p-q)] + [g(p)g(q)H(qz)h(pz)p(q-p)]dqdp$$

$$= \frac{1}{2}\int_0^1\int_0^1 g(p)g(q)\big[(p-q)(H(pz)h(qz)q - H(qz)h(pz)p)\big]dqdp$$

Next, we analyze the sign of

$$(p-q)\big[H(pz)h(qz)q - H(qz)h(pz)p\big] \tag{12}$$

which in turns determines the sign of the expression above. Suppose $p > q > 0$. (12) is ≥ 0 if and only if

$$\frac{H(pz)}{h(pz)p} \geq \frac{H(qz)}{h(qz)q}.$$

This always holds if $\frac{H(c)}{h(c)c}$ is increasing in c. This always fails if $\frac{H(c)}{h(c)c}$ is *decreasing* in c. A similar argument shows that if $q > p > 0$, an identical logic holds.

Finally, we look at the case where H has support only on $[0, \overline{c}]$. Suppose that $z > \overline{c}$. In particular, there then exists $p^* < 1$ such that $H(p^* z) = 1$ for all $p \geq p^*$ and $h(p^* z) = 0$. In this case, (12) will be < 0 for $p > p^* > q$. Moreover, as $z \to \infty$, we have that $p^* \to 0$, meaning that the integral of (12) with respect to $g(p)g(q)$ will always be < 0, meaning the contract can always be improved by substituting towards push. □

Theorem 4. *Fix a probability of at least one success $\theta \leq 1 - (1 - \mathbb{E}[p])^N$. Consider a pure pull mechanism $z(\theta)$ and a pure push mechanism $y(\theta)$ such that*

$$1 - \left(1 - \sigma(0, z(\theta))\right)^N = 1 - \left(1 - \sigma(y(\theta), 0)\right)^N = \theta$$

Pure pull is cheaper in expectation than pure push if and only if $\frac{y(\theta)}{\mathbb{E}[p]} \geq z(\theta)$. If θ is small, pure pull is cheaper in expectation. If θ is large, pure push is cheaper in expectation.

Proof. Note that σ and θ are comonotonic, and that fixing θ, σ is decreasing in N; hence, the insights with N large are identical to those where σ is sufficiently small. We write $y(\sigma) = y(\theta)$ and $z(\sigma) = z(\theta)$. Pure pull is weakly cheaper than pure push if and only if

$$\int_0^1 y(\sigma)H(y(\sigma))g(p)dp = y(\sigma)H(y(\sigma)) \geq z(\sigma)\int_0^1 pH(pz(\sigma))g(p)dp = z(\sigma)\sigma(0, z(\sigma))$$

By the hypothesis, we also have

$$\sigma(0, z(\sigma)) = \sigma(y(\sigma), 0) = \int_0^1 pH(y(\sigma))g(p)dp = \mathbb{E}[p]H(y(\sigma))$$

Meaning pure pull is weakly cheaper if and only if

$$y(\sigma)H(y(\sigma)) \geq z(\sigma)\mathbb{E}[p]H(y(\sigma)) \iff \frac{y(\sigma)}{\mathbb{E}[p]} \geq z(\sigma).$$

This gives the first result.

Next, we consider how this inequality varies with θ. Note that

$$\mathbb{E}[p]H(y(\sigma)) = \sigma \implies y'(\sigma) = \frac{1}{\mathbb{E}[p]h(y(\sigma))}$$
$$\int_0^1 pH(pz(\sigma))g(p)dp = \sigma \implies z'(\sigma) = \frac{1}{\int_0^1 p^2 g(p)h(pz(\sigma))dp}.$$

Since $y(0) = z(0) = 0$, for σ close to 0, we thus have that

$$\begin{aligned} \frac{y(\sigma)}{\mathbb{E}[p]} > z(\sigma) &\iff \frac{y'(\sigma)}{\mathbb{E}[p]} > z'(\sigma) \\ \iff \frac{1}{\mathbb{E}[p]^2 h(0)} &> \frac{1}{\int_0^1 p^2 g(p) h(0) dp} \\ &\iff \mathbb{E}[p^2] > \mathbb{E}[p]^2, \end{aligned}$$

which always holds strictly. Thus, for σ (and hence θ) small, pure pull always dominates.

For the case with θ large, we first consider the case where H has bounded support on $[0, \overline{c}]$. Suppose $\sigma \to \mathbb{E}[p]$, which is its upper bound. The expected cost of pure push to achieve this is $y(\sigma) = \overline{c}$, since d screens in all firms. However, under pure pull note that for any $z < \infty$, we necessarily have

$$\int_0^1 pg(p)H(pz)dp < \mathbb{E}[p],$$

since there are always some firms with type $p \approx 0$ who have no incentive to enter. Thus, $y(\mathbb{E}[p]) = \overline{c} < z(\mathbb{E}[p]) = \infty$, meaning pure push is cheaper.

Consider now the case where H has unbounded support, in which case both $y(\sigma)$ and $z(\sigma)$ approach ∞ as $\mathbb{E}[p] \to \infty$. Consider the function $\phi(p) = pH(pz)$. Note that

$$\phi''(p) = 2zh(pz) + pz^2 h'(pz) = zh(pz)\big(2 + pz\frac{h'(pz)}{h(pz)}\big).$$

Since h is log concave, we must have $\lim_{z\to\infty} \frac{h'(pz)}{h(pz)} < 0$, meaning that for h large, $2 + pz\frac{h'(pz)}{h(pz)} < 0$ and hence $\phi''(p)$ is concave. But then, by Jensen's inequality, for z sufficiently large, $pH(pz(\sigma))$ is concave everywhere except arbitrary close to 0 meaning:

$$\sigma = \mathbb{E}[p]H(y(\sigma)) = \int_0^1 pH(pz(\sigma))g(p)dp < \int_0^1 pg(p)dp \cdot H\Big(\int_0^1 pg(p)dpz(\sigma)\Big) = \mathbb{E}[p]H\big(\mathbb{E}[p]z(\sigma)\big)$$

We then have

$$\mathbb{E}[p]H(y(\sigma)) < \mathbb{E}[p]H\big(\mathbb{E}[p]z(\sigma)\big) \iff y(\sigma) < \mathbb{E}[p]z(\sigma),$$

which gives the result. Thus, for σ sufficiently large, pure push becomes cheaper. $\square$

Proposition 3. *Consider a contract (y, z), and suppose the associated probability of at least one success is $\theta = 1 - \big(1 - \sigma(y, z)\big)^N$. The payout scheme with the lowest fundraising cost for the designer is given by a single pot of money sized at $\frac{N \cdot \sigma(y,z) \cdot z}{\theta}$, which is split equally among all successful firms. I.e., the solution to (8) is given by $Z_n = \frac{N \cdot \sigma(y,z) \cdot z}{\theta}$ for all n.*

Proof. With slight abuse of notation, note that the probability that n other firms succeed given a contract (y, z) is:

$$\binom{N-1}{n}\sigma^n(1-\sigma)^{N-1-n}$$

where $\sigma = \sigma(y, z)$. This means that the left hand side of the constraint in (8) can be written as:

$$\begin{aligned}
\sum_{n=0}^{N-1} \frac{P_n}{n+1} Z_{n+1} &= \sum_{n=0}^{N-1} \frac{1}{n+1}\binom{N-1}{n}\sigma^n(1-\sigma)^{N-1-n} Z_{n+1} \\
&= \sum_{n=0}^{N-1} \frac{1}{n+1}\frac{(N-1)!}{(N-1-n)!n!}\sigma^n(1-\sigma)^{N-1-n} Z_{n+1} \\
&= \sum_{n=0}^{N-1} \frac{1}{N\cdot\sigma}\frac{N(N-1)!}{(N-(n+1)!)!(n+1)!}\sigma^{n+1}(1-\sigma)^{N-(n+1)} Z_{n+1} \\
&= \frac{1}{N\cdot\sigma}\sum_{n=0}^{N-1} \frac{N!}{(N!)!(n+1)!}\sigma^{n+1}(1-\sigma)^{N-(n+1)} Z_{n+1} \\
&= \frac{1}{N\cdot\sigma}\sum_{n=1}^{N}\binom{N}{n}\sigma^n(1-\sigma)^{N-n} Z_n,
\end{aligned}$$

Define $\beta_n = \binom{N}{n}\sigma^n(1-\sigma)^{N-n}$. Define $\alpha_i = \frac{\beta_i}{\sum_{i=1}^N \beta_i}$. Note that $\sum_{i=1}^N \alpha_i = 1$. Moreover, we have that $\sum_{i=1}^N \beta_i = 1-(1-\sigma)^N = \theta$, i.e. the probability of at least one success. Using this, we rewrite the problem as:

$$\min_{Z_1,\ldots,Z_n} \max_{i\in 1,\ldots,N} Z_i \text{ s.t. } \sum_{n=1}^N \alpha_n Z_n = \frac{N\cdot\sigma\cdot z}{\theta} \tag{13}$$

i.e. it is minimizing the maximum value of Z_i subject to the linear constraint $\sum_{n=1}^N \beta_n Z_n = N\cdot\sigma\cdot z$.

Let Z_m be the solution to (13). By construction, we must have

$$Z_m = \sum_{i=1}^N \alpha_n Z_m \geq \sum_{i=1}^N \alpha_n Z_n = \frac{N\cdot\sigma\cdot x}{\theta}$$

i.e. $\frac{N\cdot\sigma\cdot z}{\theta}$ is a lower bound for Z_m. But note that this lower bound can be achieved with equality by simply setting $Z_m = Z_i$ for all i. Thus, we have $Z_n = \frac{N\cdot\sigma\cdot z}{\theta}$ for all n.

□